From Regional to Global Brain: A Novel Hierarchical Spatial-Temporal Neural Network Model for EEG Emotion Recognition

Yang Li, *Student Member, IEEE*, Wenming Zheng, *Senior Member, IEEE*, Lei Wang, *Senior Member, IEEE*, Yuan Zong, *Student Member, IEEE*, and Zhen Cui, *Member, IEEE*

Abstract—In this paper, we propose a novel Electroencephalograph (EEG) emotion recognition method inspired by neuroscience with respect to the brain response to different emotions. The proposed method, denoted by R2G-STNN, consists of spatial and temporal neural network models with regional to global hierarchical feature learning process to learn discriminative spatial-temporal EEG features. To learn the spatial features, a bidirectional long short term memory (BiLSTM) network is adopted to capture the intrinsic spatial relationships of EEG electrodes within brain region and between brain regions, respectively. Considering that different brain regions play different roles in the EEG emotion recognition, a region-attention layer into the R2G-STNN model is also introduced to learn a set of weights to strengthen or weaken the contributions of brain regions. Based on the spatial feature sequences, BiLSTM is adopted to learn both regional and global spatial-temporal features and the features are fitted into a classifier layer for learning emotion-discriminative features, in which a domain discriminator working corporately with the classifier is used to decrease the domain shift between training and testing data. Finally, to evaluate the proposed method, we conduct both subject-dependent and subject-independent EEG emotion recognition experiments on SEED database, and the experimental results show that the proposed method achieves state-of-the-art performance.

Index Terms—EEG emotion recognition, regional to global, spatial-temporal network

1 Introduction

Emotion plays an essential role in human life [1]. Positive emotions could be helpful to improve the efficiency of our daily work, while negative emotions may influence our decision-making, attention, or even health [2]. Although it is easier for us to identify the other people's emotion from their facial expressions or speeches, it is still difficult for machine to do such work. Nevertheless, the study of emotion recognition using computer had attracted more and more researchers during the past several years, and emotion recognition had become a hot research topic in the research community of affective computing and pattern recognition [3]. Basically, emotion recognition methods could be categorized into the verbal behavior based methods (e.g., emotion recognition based on speech signals) and the nonverbal behavior based methods (e.g., emotion recognition based on facial expression images or physiological signals [4]). As a typical physiological signal, Electroencephalograph (EEG) had been widely applied to dealing with emotion recognition in recent years.

In dealing with EEG based emotion recognition problem, we usually encounter two major technical challenges. One is how to extract discriminative emotional feature from EEG signals, and the other one is how to develop more effective computational model for emotion recognition. Typically, EEG features can be extracted from time domain, frequency domain, and time-frequency domain [5]. For example, Lin et al. [6] investigated the relationships between emotional states and brain activities, and extracted power spectrum density, differential asymmetry power, and rational asymmetry power as the features of EEG signals. For the computational model problem, researchers have proposed many methods and models to recognize emotions through EEG signals [7], [8], [9], [10], [11]. Garca-Martnez et al. [12] summarized the most recent works that have applied nonlinear methods in EEG signal analysis to emotion recognition. Among the various EEG emotion recognition methods, it is notable that the recent development of deep learning based methods are becoming dominant for improving the performance of EEG emotion recognition. For example,

- *Y. Li is with the Key Laboratory of Child Development and Learning Science (Ministry of Education), School of Information Science and Engineering, Southeast University, Nanjing, Jiangsu 210096, China. E-mail: yang_li@seu.edu.cn.*
- *W. Zheng and Y. Zong are with the Key Laboratory of Child Development and Learning Science (Ministry of Education), School of Biological Sciences and Medical Engineering, Southeast University, Nanjing, Jiangsu 210096, China. E-mail: {wenming_zheng, xhzongyuan}@seu.edu.cn.*
- *L. Wang is with the School of Computing and Information Technology, University of Wollongong, Wollongong, NSW 2500, Australia. E-mail: leiw@uow.edu.au.*
- *Z. Cui is with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, Jiangsu 210096, China. E-mail: zhen.cui@njust.edu.cn.*

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从区域到全局大脑：一种用于脑电图情绪识别的新型层次空间-时间神经网络模型

杨立，IEEE 学生会员，郑文铭，IEEE 高级会员，王雷，IEEE 高级会员，宗元，IEEE 学生会员，崔振，IEEE 会员

摘要一本文提出了一种受神经科学启发的脑电图（EEG）情绪识别方法，该方法针对大脑对不同情绪的响应。所提出的方法，称为 R2G-STNN，由具有从区域到全局的层次特征学习过程的空间和时间神经网络模型组成，以学习有区分度的空间-时间 EEG 特征。为了学习空间特征，采用双向长短期记忆（BiLSTM）网络来捕获大脑区域内 EEG 电极的内在空间关系以及大脑区域之间的空间关系。考虑到不同的大脑区域在 EEG 情绪识别中扮演不同的角色，R2G-STNN 模型中引入了一个区域注意力层，以学习一组权重，以增强或减弱大脑区域的贡献。 基于空间特征序列，采用 BiLSTM 学习区域和全局时空特征，并将这些特征拟合到分类器层以学习情绪判别特征，其中与分类器协同工作的领域判别器用于减少训练和测试数据之间的领域偏差。最后，为评估所提出的方法，我们在 SEED 数据库上进行了受试者相关和受试者无关的脑电图情绪识别实验，实验结果表明所提出的方法达到了最先进的性能。

关键词一脑电图情绪识别，区域到全局，时空网络

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1 引言

情绪在人类生活中扮演着至关重要的角色[1]。积极情绪有助于提高我们日常工作的效率，而消极情绪可能会影响我们的决策、注意力甚至健康[2]。尽管我们更容易通过他人的面部表情或言语来识别情绪，但机器要完成这项工作仍然很困难。然而，近年来，使用计算机进行情绪识别的研究吸引了越来越多的研究人员，情绪识别已成为情感计算和模式识别研究领域的热门研究课题[3]。基本上，情绪识别方法可分为基于言语行为的方法（例如，基于语音信号的情绪识别）和基于非言语行为的方法（例如，基于面部表情图像或生理信号的情绪识别[4]）。作为典型的生理信号，脑电图（EEG）近年来已被广泛应用于情绪识别领域。

在处理基于脑电图（EEG）的情绪识别问题时，我们通常会遇到两大主要技术挑战。一是如何从脑电信号中提取具有区分性的情绪特征，二是如何开发更有效的情绪识别计算模型。通常，脑电特征可以从时域、频域和时频域中提取[5]。例如，Lin 等人[6]研究了情绪状态与大脑活动之间的关系，并提取了功率谱密度、差异对称功率和理性对称功率作为脑电信号的特征。对于计算模型问题，研究人员已经提出了许多通过脑电信号识别情绪的方法和模型[7]、[8]、[9]、[10]、[11]。Garca-Martnez 等人[12]总结了最近应用非线性方法在脑电信号分析中用于情绪识别的研究成果。在各种脑电情绪识别方法中，值得注意的是基于深度学习的方法近年来已成为主导，以提高脑电情绪识别的性能。例如，

Y. Li 任职于教育部儿童发展与学习科学重点实验室，东南大学信息科学与工程学院，江苏省南京市 210096，中国。

E-mail: yang_li@seu.edu.cn。
W. Zheng 和 Y. Zong 任职于教育部儿童发展与学习科学重点实验室，东南大学生物科学与医学工程学院，江苏省南京市 210096，中国。E-mail: {wenming_zheng, xhzongyuan}@seu.edu.cn。L. Wang 任职于伍伦贡大学计算与信息技术学院，澳大利亚新南威尔士州伍伦贡市 NSW 2500。

E-mail: leiw@uow.edu.au。
Z. Cui 是南京科技大学的计算机科学与工程学院的一员，位于江苏省南京市 210096。
E-mail: zhen.cui@njust.edu.cn。

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Zheng et al. [7] introduced deep belief networks (DBNs) to construct EEG-based emotion recognition models. Pandey et al. [13] proposed a multilayer perceptron neural network for subject-independent emotion recognition. Song et al. [14] constructed a graph relation based on multi-channel EEG data and then performed graph convolution on it to extract feature for classification. Li et al. [15] considered the domain shift for EEG data, and utilized the difference between two brain hemispheres to decrease this shift and achieved the state-of-the-art performance.

Although various EEG emotion recognition methods had been proposed in the past several years, there are still some major issues that should be well investigated in order to further improve the EEG emotion recognition performance. The first one is to extract robust high-level semantic features from the EEG signal. The current EEG recognition methods usually employ some handcraft features, such as statistic features in time domain, band power in frequency domain, and discrete wavelet transform in time-frequency domain [5]. It is desired to investigate more powerful discriminative deep features with both spatial and temporal information of EEG signals [16]. The second issue is about what brain regions are more contributive to the emotion recognition and how to make use of these topographical information of these brain regions to improve the EEG emotion recognition. The recent neuroscience researches showed that human's emotion is closely related to a variety of brain cortex regions [17], such as the orbital frontal cortex, ventral medial prefrontal cortex, amygdala [18], [19], [20]. Consequently, the EEG signals associated with different brain regions would contribute differently to emotion recognition [21], [22], [23], [24] and hence making use of the spatial information of EEG signals would be helpful for emotion recognition and could provide a physiological explanation to understand human emotion. The third issue is how to utilize the temporal information of EEG signals across different brain regions to improve emotion recognition. This is because EEG signals are dynamical time series and the temporal information usually carries important emotion messages that are very helpful to identify different emotions.

Inspired by the neuroscience finding that the brain response to different emotions would be varied in different brain regions. In this paper we propose a novel neural network model, denoted by R2G-STNN, to address the aforementioned three major issues in EEG emotion recognition. The basic idea is to integrate the EEG spatial-temporal information of both local and global brain regions into the EEG features to boost the emotion recognition performance. Specifically, R2G-STNN consists of both spatial and temporal neural network layers with regional to global (R2G) hierarchies of feature learning process to capture the emotional responses and structural relation of different brain regions for learning the discriminative spatial-temporal EEG features. The R2G-STNN framework includes the following major modules:

1) *Feature extractor*. The feature extractor module aims to learn discriminative spatial and temporal EEG features using bidirectional long short term memory (BiLSTM) network within each brain region and between the different brain regions, respectively. Considering that different brain regions play different roles in the EEG emotion recognition, a region-attention layer is also introduced to learn a set of weights indicating the contributions of brain regions.
2) *Classifier and discriminator*. The classifier module aims to predict the emotion class information based on the spatial-temporal features obtained from the feature extractor module. It can also guide the overall neural network learning towards generating more discriminative EEG features for emotion classification. Moreover, we also introduce a discriminator to alleviate the domain shift between source and target domain data, which will enable the hierarchical feature leaning process to generate emotion discriminative but domain adaptive EEG features.

By combining the aforementioned three modules together, we can learn more discriminative and domain-robust EEG features for improving emotion recognition performance. In summary, the major contributions of this paper include the following two major parts:

- Propose a novel neuroscience inspired hierarchical spatial-temporal EEG feature learning model, which is able to capture both spatial and temporal emotion information from EEG signals within each brain region and across different brain regions;
- Propose a weighting method to evaluate the different contributions of the brain regions to the EEG emotion recognition, which would be advantageous to further improve the EEG emotion recognition by enhancing the influence of the most contributive brain regions while alleviating the influence of the less contributive regions through the region weights.

The remainder of this paper is organized as follows: In Section 2, we overview the preliminary work of bidirectional long short term memory network. In Section 3, we specify the method of R2G-STNN as well as its application to EEG emotion recognition. In Section 4, we conduct extensive experiments to evaluate the proposed method for EEG emotion recognition. Finally, in Sections 5, we discuss and conclude the paper.

2 Preliminary

In this section, we will briefly overview the preliminary work of bidirectional long short term memory network [25] and then address how we can apply it to the EEG feature extraction task.

The conventional single directional long short term memory (LSTM) network [26] is a special type of recurrent neural network (RNN) [27], which usually consists of three gates and a cell state for dealing with the long-term dependence of data sample sequence. These gates allow LSTM to keep the important data information and forget unnecessary information of the data samples [28]. However, one shortcoming of conventional LSTM is that it only make use of the previous context. The BiLSTM module is able to process data using two directions with separate hidden layers, respectively [25]. As a result, compared with the traditional LSTM model, BiLSTM can access the long-range context in both input directions, and hence it could be better used to model time sequences.

郑等[7]将深度信念网络（DBNs）引入构建基于脑电图（EEG）的情绪识别模型。潘等[13]提出了一种多层感知器神经网络用于进行与被试无关的情绪识别。宋等[14]基于多通道脑电图数据构建了图关系，然后对其进行图卷积以提取用于分类的特征。李等[15]考虑了脑电图数据的领域偏移，并利用两个大脑半球之间的差异来减少这种偏移，实现了最先进的性能。

尽管在过去几年中已经提出了各种脑电图（EEG）情绪识别方法，但仍有一些重大问题需要深入调查，以进一步提高 EEG 情绪识别性能。第一个问题是从 EEG 信号中提取鲁棒的高级语义特征。当前的 EEG 识别方法通常采用手工特征，例如时域统计特征、频域的频带功率以及时频域的小波变换[5]。人们期望研究具有 EEG 信号时空信息的更强大的判别性深度特征[16]。第二个问题是哪些脑区对情绪识别贡献更大，以及如何利用这些脑区的拓扑信息来提高 EEG 情绪识别。最近神经科学研究表明，人类的情绪与多种脑皮层区域密切相关[17]，例如眶额叶皮层、腹内侧前额叶皮层、杏仁核[18]、[19]、[20]。因此，与不同脑区相关的脑电信号在情绪识别中的贡献会有所不同[21], [22], [23], [24]，因此利用脑电信号的空间信息有助于情绪识别，并能为理解人类情绪提供生理学解释。第三个问题是如何利用不同脑区脑电信号的时间信息来提高情绪识别。这是因为脑电信号是动态时间序列，时间信息通常包含重要的情绪信息，对识别不同情绪非常有帮助。

受神经科学研究发现大脑对不同情绪的反应在不同脑区会有所不同。在本文中，我们提出了一种新型神经网络模型，记为 R2G-STNN，以解决脑电图情绪识别中的上述三大主要问题。其基本思想是将局部和全局脑区的脑电图时空信息整合到脑电图特征中，以提高情绪识别性能。具体而言，R2G-STNN 由具有区域到全局（R2G）层次特征学习过程的时空神经网络层组成，以捕捉不同脑区的情绪反应和结构关系，从而学习具有判别性的时空脑电图特征。R2G-STNN 框架包括以下主要模块：

1) 特征提取器。特征提取器模块旨在使用双向长短期记忆（BiLSTM）网络分别在每个脑区内以及不同脑区之间学习具有判别性的时空脑电图特征。考虑到不同脑区在脑电图情绪识别中扮演不同角色，引入了区域注意力层来学习一组权重，用以表示脑区的贡献。

2) 分类器和判别器。分类器模块旨在基于特征提取器模块获得的时空特征来预测情绪类别信息。它还可以引导整个神经网络学习，以生成更具判别性的脑电图特征用于情绪分类。此外，我们还引入了判别器来缓解源域和目标域数据之间的领域偏移，这将使分层特征学习过程能够生成具有情绪判别性但领域自适应的脑电图特征。

通过将上述三个模块结合起来，我们可以学习到更具判别性和领域鲁棒的脑电图特征，以提升情绪识别性能。总之，本文的主要贡献包括以下两个方面：

提出一种受神经科学启发的层次化时空 EEG 特征学习模型，该模型能够从每个脑区的 EEG 信号中以及跨不同脑区捕获空间和时间情绪信息；提出一种权重方法来评估不同脑区对 EEG 情绪识别的贡献，通过区域权重增强最具贡献脑区的影响，同时减轻贡献较小的脑区的影响，从而进一步改进 EEG 情绪识别。本文其余部分组织如下：在第二节中，我们概述了双向长短期记忆网络的基础工作。在第三节中，我们详细说明了 R2G-STNN 的方法及其在 EEG 情绪识别中的应用。在第四节中，我们进行了广泛的实验来评估所提出的方法用于 EEG 情绪识别。最后，在第五节中，我们讨论并总结了本文。

2 初步

在本节中，我们将简要概述双向长短期记忆网络[25]的初步工作，然后探讨如何将其应用于脑电图特征提取任务。

传统的单向长短期记忆（LSTM）网络[26]是一种特殊的循环神经网络（RNN）[27]，通常由三个门和一个细胞状态组成，用于处理数据样本序列的长期依赖关系。这些门允许 LSTM 保留重要的数据信息并遗忘数据样本中不必要的信息[28]。然而，传统 LSTM 的一个缺点是它只利用之前的上下文。BiLSTM 模块能够使用两个独立的隐藏层分别以两个方向处理数据[25]。因此，与传统的 LSTM 模型相比，BiLSTM 可以在两个输入方向上访问长距离上下文，因此它更适合用于建模时间序列。

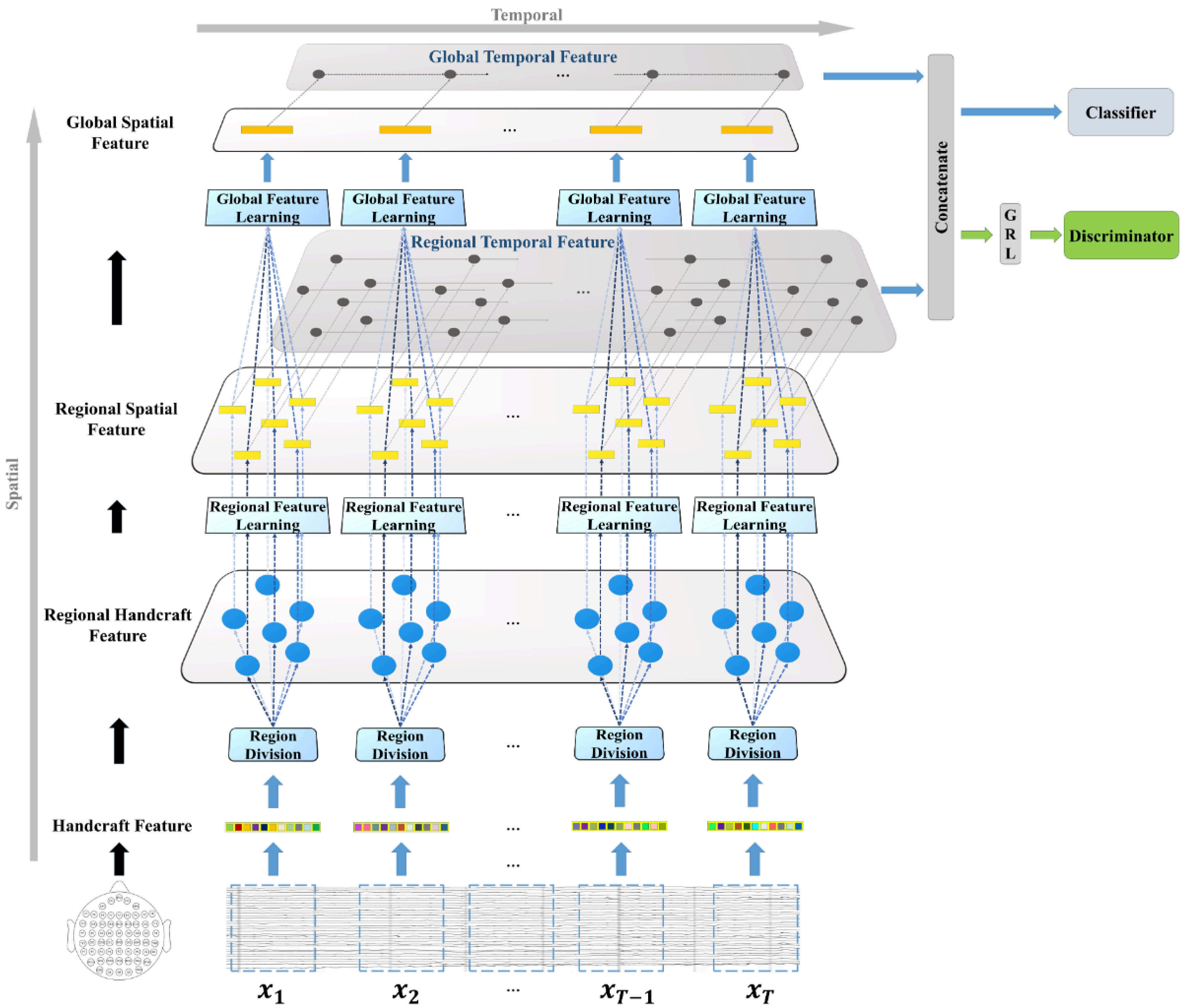

Fig. 1. The framework of R2G-STNN. The region to global (R2G) process includes two directions, i.e., spatial and temporal streams. The spatial stream constructs the relation in and among all the brain regions hierarchically, while the temporal stream captures the EEG signal's dynamic information as well as learning from the brain regions' time sequences.

When dealing with the EEG data processing problem, it is interesting to see that the EEG signals associated with the electrodes of each brain region can be treated as virtual sequence since the dimensions of all the electrodes are same. For this reason, the EEG data can be fed into BiLSTM module to extract high-level deep features, which will contain the spatial relation information. Moreover, benefiting from the less probable disturbance of electrode arrangement in the input sequence, we use BiLSTM rather than conventional single directional LSTM to model the electrodes' data in each brain region, then forward construct their spatial relations.

3 R2G-STNN for EEG-Based Emotion Recognition

In this section, we will specify the R2G-STNN model as well as the method of applying this model to dealing with EEG emotion recognition. Fig. 1 illustrates the framework of the R2G-STNN method, from which we can see that the R2G-STNN method consists of three major parts, i.e., feature extractor part, the classifier part, and the discriminator part. In what follows, we will address these parts in details.

3.1 Spatial Feature Learning

Suppose that we are given a trial of raw EEG signals $\mathbf{S}$. Then, we divide $\mathbf{S}$ into several segments. For each segment of EEG signals, we extract a set of handcraft features (e.g., the differential entropy (DE) features [7] on five EEG frequency bands, i.e., δ band (1-3 Hz), θ band (4-7 Hz), α band (8-13 Hz), β band (14-30 Hz), γ band (31-50 Hz)) from each EEG electrode. Moreover, to explore the dynamic temporal information of EEG signals, every $T = 9$ neighboring EEG segments are chosen to constitute an EEG sample. In this case, each EEG sample will correspond to a handcraft feature tensor.

Let $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T] \in \mathbb{R}^{d\times n\times T}$ denote an EEG sample, where $\mathbf{x}_i$ denotes a handcraft feature matrix extracted from the ith segment of EEG signals (denoted by blue dashed rectangle box shown at the bottom of Fig. 1), d, n and T denote the number of features associated with each EEG electrode, the number of electrodes, and the number of EEG segments in one EEG sample, respectively. Then, from Fig. 1 we can see that, for each $\mathbf{x}_i$ $(i \in \{1, 2, \ldots, T\})$, the spatial feature learning for $\mathbf{x}_i$ consists of two feature extracting layers, which achieves a progressive feature learning from regional brain to global brain.

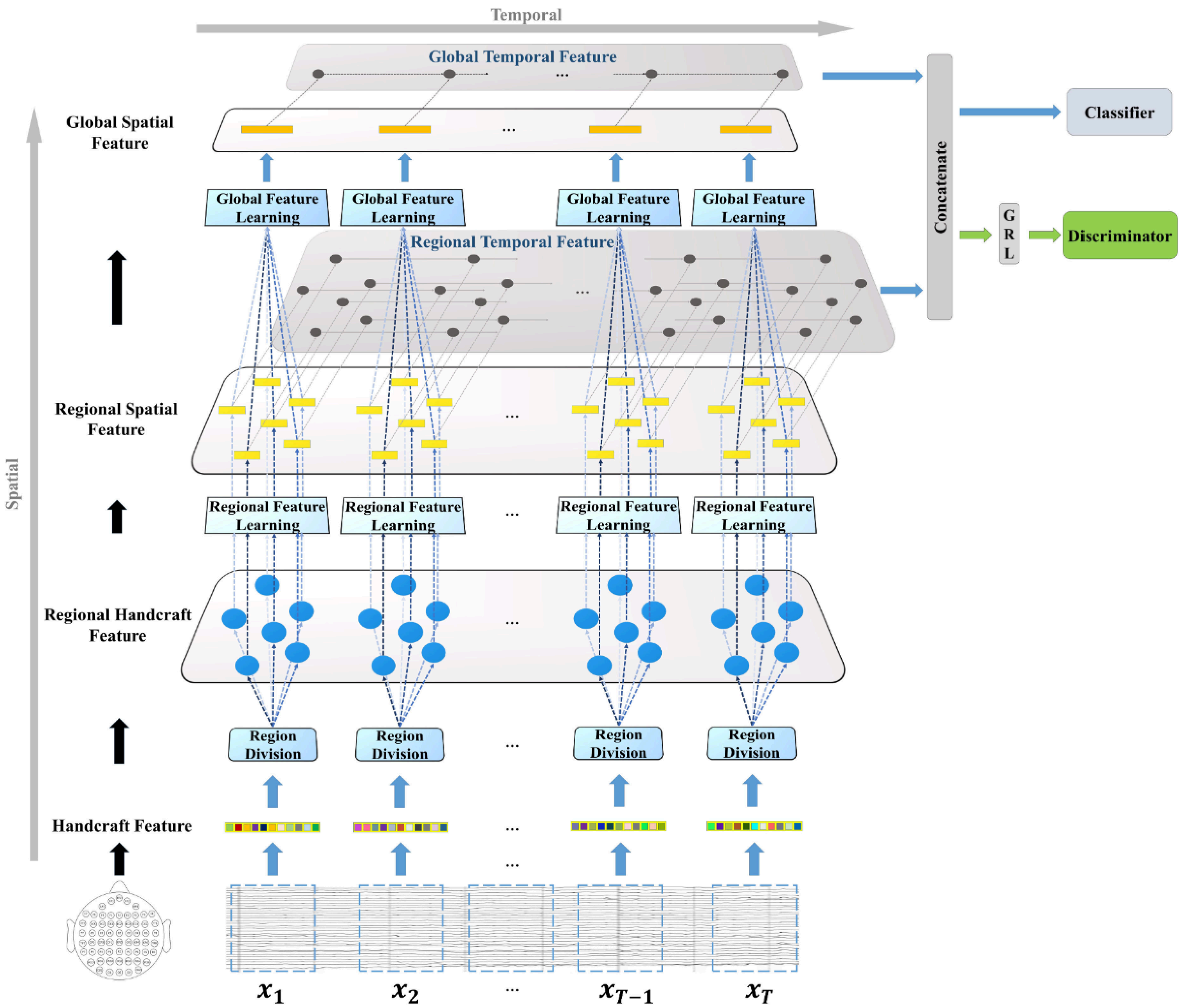

图 1. R2G-STNN 的框架。区域到全局（R2G）过程包括两个方向，即空间流和时间流。空间流逐层构建所有大脑区域之间的关系，而时间流捕获脑电图信号的动态信息，并从大脑区域的时间序列中学习。

在处理脑电图数据处理问题时，有趣的是，每个脑区电极相关的脑电图信号可以被视为虚拟序列，因为所有电极的维度相同。因此，脑电图数据可以输入到 BiLSTM 模块中提取高级深度特征，这些特征将包含空间关系信息。此外，由于输入序列中电极排列的干扰可能性较低，我们使用 BiLSTM 而不是传统的单向 LSTM 来对每个脑区的电极数据进行建模，然后正向构建它们的空间关系。

3 R2G-STNN 用于基于脑电图的情感识别
识别

在本节中，我们将具体说明 R2G-STNN 模型以及将该模型应用于脑电图情绪识别的方法。图 1 展示了 R2G-STNN 方法的框架，从中我们可以看到 R2G-STNN 方法由三个主要部分组成，即特征提取部分、分类器部分和判别器部分。

接下来，我们将详细阐述这些部分。

3.1 空间特征学习

假设我们给定一组原始脑电图信号 S。然后，我们将 S 分成若干段。对于每段脑电图信号，我们从每个脑电图电极提取一组手工特征（例如，在五个脑电图频段上的微分熵（DE）特征[7]，即 d 带（1-3 Hz）、u 带（4-7 Hz）、a 带（8-13 Hz）、b 带（14-30 Hz）、g 带（31-50 Hz））。此外，为了探索脑电图信号的动态时间信息，每 9 个相邻的脑电图段被选出来构成一个脑电图样本。在这种情况下，每个脑电图样本将对应一个手工特征张量。

令 $X = \{x_1, x_2, . . . , x\_T\} \in \mathbb{R}$ 表示一个脑电图样本，其中 x 表示从第 i 个脑电信号段（如图 1 底部蓝色虚线矩形框所示）中提取的手工特征矩阵，d、n 和 T 分别表示每个脑电电极关联的特征数量、电极数量以及一个脑电样本中的脑电段数量。然后，从图 1 中我们可以看到，对于每个 x(i ∈ {1, 2, . . . , T})，x 的空间特征学习包括两个特征提取层，实现了从局部大脑到全局大脑的渐进式特征学习。

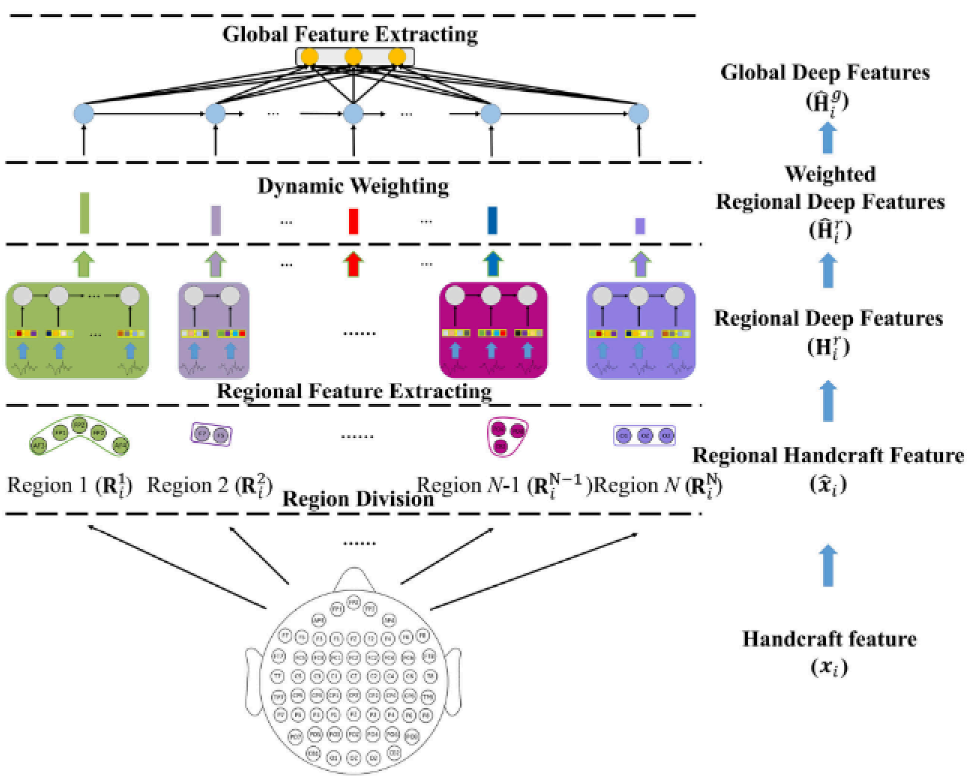

Fig. 2. The regional to global spatial feature learning process, which consists of three layers, i.e., regional feature learning layer, dynamic weight layer, and global feature learning layer.

Fig. 2 illustrates the detailed learning procedure for the input handcraft feature matrix $\mathbf{x}_i$. First, we group the columns of $\mathbf{x}_i$ into several clusters, where each cluster corresponds to a brain region determined by the spatial locations of EEG electrodes, resulting in a set of regional handcraft feature vectors in each brain region. Then, the regional handcraft feature vectors of each brain region are fed into the same amount of BiLSTM networks to learn regional deep features. After extracting the regional spatial feature, a region-attention layer is used to learn a set of weights indicating the importance of the divided regions and the weights are used to penalize the regional deep features. Finally, at the top of the spatial feature learning layer, the weighted regional feature vectors are further fed into BiLSTM to learn global deep features. These detailed operations of spatial feature learning are described as follows:

3.1.1 Regional Feature Learning

Let $\mathbf{v}_{ij}$ denote the handcraft feature vector associated with the jth EEG electrode. Let $\mathbf{x}_i = [\mathbf{v}_{i1}, \mathbf{v}_{i2}, \ldots, \mathbf{v}_{in}] \in \mathbb{R}^{d \times n}$. Now we group the n columns of $\mathbf{x}_i$ into several clusters according to the associated electrodes, where each cluster corresponds to a brain region, e.g.,

$$\text{brain region 1:} \quad \mathbf{R}_i^1 = [\mathbf{v}_{i1}^1, \mathbf{v}_{i2}^1, \ldots, \mathbf{v}_{in_1}^1],$$
$$\cdots$$
$$\text{brain region N:} \quad \mathbf{R}_i^N = [\mathbf{v}_{i1}^N, \mathbf{v}_{i2}^N, \ldots, \mathbf{v}_{in_N}^N],$$

where N is the number of brain regions, n_j denotes the number of electrodes in the jth brain region, $n_1 + \cdots + n_N = n$.

In this case, we change the order of the columns of $\mathbf{x}_i$ and express it as a block matrix denoted by

$$\hat{\mathbf{x}}_i = [\mathbf{R}_i^1, \ldots, \mathbf{R}_i^N],$$

which is called the regional handcraft feature matrix. For each block matrix $\mathbf{R}_i^j$ $(j = 1, \ldots, N)$, it is notable that each column corresponds to an EEG electrode. Consequently, we can model the spatial relationships of the electrodes by applying BiLSTM to the columns of each block matrix to obtain high-level deep features, which can be formulated as

$$\mathcal{L}(\mathbf{R}_i^1) = [\mathbf{h}_{i1}^1, \mathbf{h}_{i2}^1, \ldots, \mathbf{h}_{in_1}^1] \in \mathbb{R}^{2d_r \times n_1}, \tag{1}$$
$$\cdots$$
$$\mathcal{L}(\mathbf{R}_i^N) = [\mathbf{h}_{i1}^N, \mathbf{h}_{i2}^N, \ldots, \mathbf{h}_{in_N}^N] \in \mathbb{R}^{2d_r \times n_N}, \tag{2}$$

where $\mathcal{L}(\cdot)$ denotes the BiLSTM operation, $\mathbf{h}_{ik}^j \in \mathbb{R}^{2d_r}$ denotes the kth forward and backward hidden units of BiLSTM, d_r is the hidden units dimension of regional spatial BiLSTM. After the above regional feature learning process, we concatenate the last hidden units of BiLSTM as the feature representation. In this case, the local features of all the regions can be expressed as

$$\mathbf{H}_i^r = [\mathbf{h}_{in_1}^1, \ldots, \mathbf{h}_{in_N}^N] \in \mathbb{R}^{2d_r \times N}. \tag{3}$$

3.1.2 Attention-Based Brain Region Weights Learning

The research of neuroscience indicated that different emotions are closely related to different brain regions such as orbital frontal cortex or ventral medial prefrontal cortex. Hence, the EEG signals associated with different brain regions would contribute differently to emotion recognition. For this reason, we introduce a weighting layer to emphasize the contributions of the electrodes of the brain regions in EEG emotion recognition.

Specifically, for each brain region, we introduce a dynamic weight, denoted by matrix $\mathbf{W} = \{w_{ij}\}$, to weight electrodes of each brain region, i.e.,

$$\hat{\mathbf{H}}_i^r = \mathbf{H}_i^r \mathbf{W}, \tag{4}$$

$$\mathbf{W} = (\mathbf{Q} \tanh(\mathbf{P}\mathbf{H}_i^r + \mathbf{b}^r \mathbf{e}^T))^{\mathrm{T}}, \tag{5}$$

$$w_{ij} = \frac{\exp(w_{ij})}{\sum_{k=1}^{N} \exp(w_{kj})}, \tag{6}$$

where $\mathbf{P}$ and $\mathbf{Q}$ are learnable transformation matrices, $\mathbf{b}^r$ is the bias, and $\mathbf{e}$ denotes an N-dimensional vector with all elements 1, i.e., $\mathbf{e} = [1, 1, \ldots, 1]^T$. The columns of $\mathbf{W}$ are normalized and the elements are restricted to be nonnegative using Eq. (6). In this case, we obtain that the larger w_{ij} is, the more important ith region is.

3.1.3 Global Feature Learning

This layer is used to capture the potential structural information based on the above obtained weighted regional spatial features $\hat{\mathbf{H}}_i^r$. Here, we employ another BiLSTM network to capture the global structure of all EEG brain regions, which can be formulated as

$$\mathcal{L}(\hat{\mathbf{H}}_i^r) = [\mathbf{h}_{i1}^g, \mathbf{h}_{i2}^g, \ldots, \mathbf{h}_{iN}^g] \in \mathbb{R}^{d_g \times N}, \tag{7}$$

where $\mathbf{h}_{ik}^g$ denote the kth hidden unit of BiLSTM, d_g is dimension of global spatial BiLSTM.

For the feature vector sequence $\mathbf{h}_{i1}^g, \ldots, \mathbf{h}_{iN}^g$, we use them as input data and learn another feature vector sequence according to the following rule:

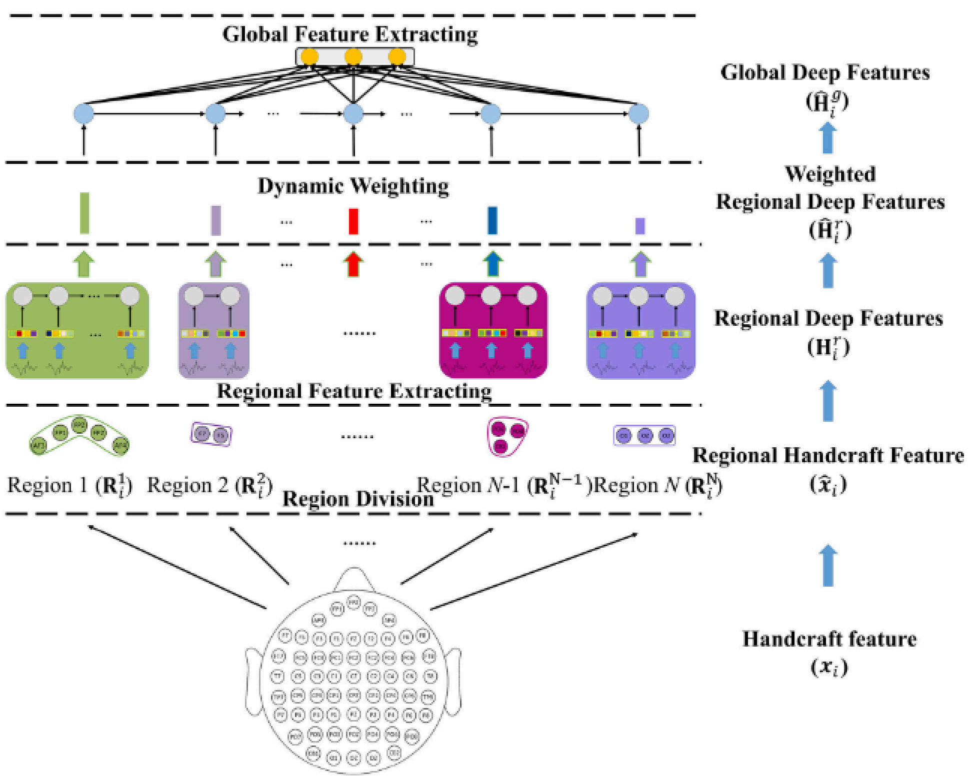

图 2. 区域到全局空间特征学习过程，该过程由三层组成，即区域特征学习层、动态权重层和全局特征学习层。

图 2 展示了输入手工特征矩阵 x 的详细学习过程。首先，我们将 x 的列分组成多个簇，每个簇对应由脑电图电极的空间位置确定的脑区，从而在每个脑区得到一组区域手工特征向量。然后，将每个脑区的区域手工特征向量输入相同数量的 BiLSTM 网络以学习区域深度特征。在提取区域空间特征后，使用区域注意力层学习一组表示划分区域重要性的权重，并将这些权重用于惩罚区域深度特征。最后，在空间特征学习层的顶部，将加权区域特征向量进一步输入 BiLSTM 以学习全局深度特征。空间特征学习的详细操作描述如下：

3.1.1 区域特征学习

设 v 表示与第 j 个脑电图电极相关的手工特征向量。设 x = (v; v; . . . ; v) ∈ R。现在我们根据相关的电极将 x 的 n 列分组为若干个簇，其中每个簇对应一个脑区，例如：

脑区 1: R = (v; v; . . . ; v);

脑区 N: R = (v; v; . . . ; v);

其中 N 是脑区的数量，n 表示第 j 个脑区中的电极数量，nþ þ n¼ n。

在这种情况下，我们改变 x 的列的顺序，并将其表示为一个分块矩阵，记为

x̂¼ ½R; . . . ; RŠ;

这被称为区域手工特征矩阵。对于每个分块矩阵 R(j ¼ 1; . . . ; N)，值得注意的是每一列对应一个脑电图电极。因此

我们可以通过将 BiLSTM 应用于每个块矩阵的列来对电极的空间关系进行建模，以获得高级深度特征，这可以表示为

LðRÞ ¼ ½h; h; . . . ; hŠ 2 R; (1)

LðRÞ ¼½h; h; . . . ; hŠ 2 R; (2)

其中 LðÞ表示 BiLSTM 操作，h2 R 表示 BiLSTM 的第 k 个正向和反向隐藏单元，dis 表示区域空间 BiLSTM 的隐藏单元维度。在上述区域特征学习过程之后，我们将 BiLSTM 的最后一个隐藏单元连接作为特征表示。在这种情况下，所有区域的地方特征可以表示为

H¼ ½h; . . . ; hŠ 2 R: (3)

3.1.2 基于注意力机制的脑区权重学习

神经科学研究表明，不同的情绪与不同的脑区（如眶额叶皮层或腹内侧前额叶皮层）密切相关。因此，与不同脑区相关的脑电信号在情绪识别中的贡献会有所不同。为此，我们引入一个权重层，以强调脑区电极在脑电情绪识别中的贡献。

具体来说，对于每个脑区，我们引入一个动态权重，记为矩阵 W ¼ fwg，用于对每个脑区的电极进行加权，即

Ĥ¼ HW; (4)

W ¼ ðQ tanhðPHþ beÞÞ; (5)

w¼ expðwÞ / k¼1 expðwÞ; (6)

其中 P 和 Q 是可学习的转换矩阵，b 是偏差，而 e 表示一个所有元素为 1 的 N 维向量，即 e = [1; 1; ...; 1]。W 的列被归一化，并且其元素通过公式(6)被限制为非负。在这种情况下，我们得到 w_i 越大，第 i 个区域就越重要。

3.1.3 全局特征学习

该层用于基于上述获得的加权区域空间特征^H 捕获潜在的结构信息。这里，我们采用另一个 BiLSTM 网络来捕获所有 EEG 脑区的全局结构，可以表示为

L(^H) = [h; h; ...; h] ∈ R (7)

其中，h 表示 BiLSTM 的第 k 个隐藏单元，d 表示全局空间 BiLSTM 的维度。

对于特征向量序列 h₁, ..., hₙ，我们将其作为输入数据，并根据以下规则学习另一个特征向量序列：

$$\hat{\mathbf{h}}_{ik}^{g} = \sigma\left(\sum_{i=1}^{N} G_{jk}^{g}\mathbf{h}_{ij}^{g} + \mathbf{b}^{g}\right), \quad k = 1, 2, \ldots, K, \tag{8}$$

$$\hat{\mathbf{H}}_{i}^{g} = [\hat{\mathbf{h}}_{i1}^{g}, \hat{\mathbf{h}}_{i2}^{g}, \ldots, \hat{\mathbf{h}}_{iK}^{g}], \tag{9}$$

where $\mathbf{G}^{g} = [G_{jk}^{g}]_{N\times K}$ is a project matrix, $\mathbf{b}^{g}$ is a bias and K is the length of the compressed sequence, $\sigma(\cdot)$ denotes a nonlinear mapping function. In this case, we finally get the global-level features associated with the ith EEG segmentation handcraft feature matrix $\mathbf{x}_i$.

3.2 Temporal Feature Learning

Let $\mathbf{h}_{i}^{j}$ $(j = 1, \ldots, N)$ denote the last hidden unit of the jth brain region associated with the ith EEG handcraft feature matrix $\mathbf{x}_i$. Let

$$\mathbf{H}^{r1} \triangleq [\mathbf{h}_{1}^{1}, \mathbf{h}_{2}^{1}, \ldots, \mathbf{h}_{T}^{1}], \tag{10}$$

$$\cdots$$

$$\mathbf{H}^{rN} \triangleq [\mathbf{h}_{1}^{N}, \mathbf{h}_{2}^{N}, \ldots, \mathbf{h}_{T}^{N}]. \tag{11}$$

Then, the columns of the feature matrix $\mathbf{H}^{rj}$ $(j = 1, \ldots, N)$ constitute a feature vector sequence associated with the jth brain region. Hence, we can apply BiLSTM to learning temporal information among the feature vector sequence, which results in the following regional temporal feature vectors:

$$\begin{aligned}\mathbf{Y}^{rt} &= [\mathcal{L}(\mathbf{H}^{r1}), \ldots, \mathcal{L}(\mathbf{H}^{rN})] \\ &= [(\mathbf{y}_{11}^{rt}, \ldots, \mathbf{y}_{1T}^{rt}), \ldots, (\mathbf{y}_{N1}^{rt}, \ldots, \mathbf{y}_{NT}^{rt})] \\ &= [\mathbf{Y}_{1}^{rt}, \ldots, \mathbf{Y}_{N}^{rt}],\end{aligned} \tag{12}$$

where $\mathbf{Y}_{j}^{rt} = [\mathbf{y}_{j1}^{rt}, \ldots, \mathbf{y}_{jT}^{rt}] \in \mathbb{R}^{d_{rt}\times T}$ denotes the regional temporal feature matrix associated with the jth brain region, d_{rt} is the hidden units dimension of regional temporal BiLSTM.

On the other hand, to learn the temporal information based on the global feature matrices $\hat{\mathbf{H}}_{i}^{g}$, we concatenate the columns of $\hat{\mathbf{H}}_{i}^{g}$ into a longer feature vector, denoted by $\hat{\mathbf{h}}_{i}^{g}$, i.e.,

$$\hat{\mathbf{h}}_{i}^{g} = [(\hat{\mathbf{h}}_{i1}^{g})^{\mathrm{T}}, \ldots, (\hat{\mathbf{h}}_{iK}^{g})^{\mathrm{T}}]^{\mathrm{T}}. \tag{13}$$

Let $\mathbf{Y}^{g} = [\hat{\mathbf{h}}_{1}^{g}, \ldots, \hat{\mathbf{h}}_{T}^{g}]$. Then, the global temporal feature $\mathbf{Y}^{gt}$ can be computed as

$$\mathbf{Y}^{gt} = \mathcal{L}(\mathbf{Y}^{g}) = [\mathbf{y}_{1}^{gt}, \ldots, \mathbf{y}_{T}^{gt}] \in \mathbb{R}^{d_{gt}\times T}, \tag{14}$$

where d_{gt} is the hidden units dimension of global temporal BiLSTM.

The final feature vector, denoted by $\mathbf{y}^{rg}$, of the EEG sample $\mathbf{X}$ (consisting of T segments) containing both regional and global information is expressed as

$$\mathbf{y}^{rg} = [(\mathbf{y}_{1T}^{rt})^{\mathrm{T}}, (\mathbf{y}_{2T}^{rt})^{\mathrm{T}}, \ldots, (\mathbf{y}_{NT}^{rt})^{\mathrm{T}}, (\mathbf{y}_{T}^{gt})^{\mathrm{T}}]^{\mathrm{T}}. \tag{15}$$

3.3 Classifier and Discriminator

Based on the final feature vector $\mathbf{y}^{rg}$, we can predict the class label of the input EEG sample $\mathbf{X}$ by using the simple linear transform approach, which can be formulated as

$$\mathbf{O} = \mathbf{G}\mathbf{y}^{rg} + \mathbf{b}_c = [o_1, o_2, \ldots, o_C], \tag{16}$$

where $\mathbf{G}$ and $\mathbf{b}_c$ are respectively the transform matrix and bias, C is the number of class. The elements of the output $\mathbf{O}$ are then fed into a softmax function for emotion class prediction, i.e.,

$$P(c|\mathbf{X}) = \max\left\{\frac{\exp(o_k)}{\sum_{i=1}^{C}\exp(o_i)} | k = 1, \ldots, C\right\}, \tag{17}$$

where $P(c|\mathbf{X})$ denotes the probability for the input $\mathbf{X}$ being predicted as the cth class.

Now suppose that we have M data samples from source domain, which are expressed as M matrices $\mathbf{X}_{i}^{S}$ $(i = 1, \ldots, M)$. Then, the loss function of the classifier can be formulated as

$$L_c(\mathbf{X}_{1}^{S}, \ldots, \mathbf{X}_{M}^{S}; \theta_f, \theta_c) = \sum_{i=1}^{M}\sum_{c=1}^{C} -\tau(l_i, c) \times \log P(c|\mathbf{X}_{i}^{S}), \tag{18}$$

where l_i denotes the ground-truth label of $\mathbf{X}_{i}^{S}$, θ_f and θ_c denote the parameters of feature extractor and classifier, and $\tau(l_i, c)$ is expressed as

$$\tau(l_i, c) = \begin{cases} 1, & \text{if } l_i = c, \\ 0, & \text{otherwise.} \end{cases} \tag{19}$$

Consequently, from (18) and (19), by minimizing the loss function $L_c(\mathbf{X}_{1}^{S} \ldots, \mathbf{X}_{M}^{S}; \theta_f, \theta_c)$, we would be able to achieve the maximal probability of correctly predicting the emotion class of each training sample.

Let $\mathbf{X}_{test}$ be a test sample. Then, we use the following formula to determine the emotion class label of $\mathbf{X}_{test}$:

$$l_{test} = \arg\max_{c}\{P(c|\mathbf{X}_{test})|c = 1, \ldots, C\}, \tag{20}$$

where l_{test} is the predicted label of the testing sample $\mathbf{X}_{test}$.

In dealing with EEG emotion recognition, it is notable that the training and testing EEG data samples may come from different domains, e.g., the training and testing data samples come from different subjects. In this case, the emotion recognition model trained based on the training data may not be well suitable for the testing data. To solve this problem, we introduce a discriminator that works corporately with the classifier to produce emotion-discriminative and domain-invariable features.

Specifically, suppose that we are given two data sets $\mathbf{X}^{S} = \{\mathbf{X}_{1}^{S}, \ldots, \mathbf{X}_{M_1}^{S}\}$ and $\mathbf{X}^{T} = \{\mathbf{X}_{1}^{T}, \ldots, \mathbf{X}_{M_2}^{T}\}$ from source domain and target domain, respectively, where M_1 and M_2 denote the number of source data set and target data set, respectively. To alleviate the domain difference, we introduce the following loss function

$$L_d(\mathbf{X}_{i}^{S}, \mathbf{X}_{j}^{T}; \theta_f, \theta_d) = -\sum_{i=1}^{M_1}\log P(0|\mathbf{X}_{i}^{S}) - \sum_{j=1}^{M_2}\log P(1|\mathbf{X}_{j}^{T}), \tag{21}$$

where $P(0|\mathbf{X}_{i}^{S})$ denotes the probability that $\mathbf{X}_{i}^{S}$ belongs to source domain while $P(1|\mathbf{X}_{i}^{T})$ denotes the probability that $\mathbf{X}_{i}^{T}$ belongs to target domain, respectively, θ_d denotes the parameter of discriminator. By maximizing the above loss function of discriminator, the feature extracting process would results in domain-invariable features to alleviate the domain difference in emotion recognition.

ĥ^¼ s X G hþ b ! ; k ¼ 1; 2; . . . ; K; (8)
i¼1

Ĥ^¼ ½^h; ^h; . . . ; ^hŠ; (9)

其中 G¼ ½G Š 是一个投影矩阵，b 是偏置，K 是压缩序列的长度，sðÞ 表示一个非线性映射函数。在这种情况下，我们最终得到与第 i 个 EEG 分割手工特征矩阵 x 相关的全局级特征。

3.2 时序特征学习

令 h(j = 1; . . . ; N) 表示与第 i 个脑电手工特征矩阵 x 相关的第 j 个脑区的最后一个隐藏单元。令

H, ½h; h; . . . ; hŠ; (10)

H, ½h; h; . . . ; hŠ: (11)

然后，特征矩阵 H(j ¼ 1; . . . ; N)的列构成与第 j 个脑区相关的特征向量序列。因此，我们可以应用 BiLSTM 来学习特征向量序列中的时间信息，从而得到以下区域时间特征向量：

Y¼ ½LðHÞ; . . . ; LðHÞŠ
¼ ½ðy; . . . ; yÞ; . . . ; ðy; . . . ; yÞŠ (12)
¼ ½Y; . . . ; YŠ;

其中 Y¼ ½y; . . . ; yŠ 2 R 表示与第 j 个脑区相关的区域时间特征矩阵，d 是区域时间 BiLSTM 的隐藏单元维度。

另一方面，为了基于全局特征矩阵^H 学习时间信息，我们将^H 的列连接成一个更长的特征向量，记为^h，即：

ĥ^¼ ½ð^hÞ; . . . ; ð^hÞŠ: (13)

令 Y = (y^1, ..., y^h)。然后，全局时间特征 Y 可以表示为

$$Y = L(Y) = (y, ..., y) \in R; \quad (14)$$

其中 d 是全局时间 BiLSTM 的隐藏单元维度。

包含区域和全局信息的 EEG 样本 X（由 T 个片段组成）的最终特征向量 y 表示为

y¼ ½ðyÞ; ðyÞ; . . . ; ðyÞ; ðyÞŠ: (15)

3.3 分类器与判别器

基于最终特征向量 y，我们可以通过简单的线性变换方法预测输入的 EEG 样本 X 的类别标签，这可以表示为

O ¼ Gyþ b¼ ½o; o; . . . ; o Š; (16)

G 和 b 分别表示变换矩阵和偏置，C 是类别数

输出 O 的元素随后被输入到 softmax 函数中进行情绪类别预测，即

P ðcjXÞ ¼ max (expðo Þ / P i¼1 expðo Þ jk ¼ 1; . . . ; C); (17)

其中 P ðcjXÞ表示输入 X 被预测为第 c 类的概率。

现在假设我们有来自源域的 M 个数据样本，这些样本表示为 M 个矩阵 X(i ¼ 1; . . . ; M)。那么，分类器的损失函数可以表示为

LðX; . . . ; X; u; uÞ ¼ XM i¼1 XC c¼1 tðl ; cÞ logP ðcjXÞ;

(18) 其中 l 表示 X 的真实标签，u 和 u 分别表示特征提取器和分类器的参数，tðl ; cÞ表示为

tðl; cÞ ¼ 1; 如果 l = c; 0; 否则: (19)

因此，从(18)和(19)式可以看出，通过最小化损失函数 L(X, ..., X; u, u)，我们可以实现正确预测每个训练样本情绪类别的最大概率。

设 X 为一个测试样本。然后，我们使用以下公式来确定 X 的情绪类别标签：

l ¼ arg max c f P ðcjXÞjc ¼ 1; . . . ; C g; (20)

其中 l 是测试样本 X 的预测标签。

在处理脑电图情绪识别时，值得注意的是训练和测试的脑电图数据样本可能来自不同的领域，例如训练和测试数据样本来自不同的被试。在这种情况下，基于训练数据训练的情绪识别模型可能并不适合测试数据。为了解决这个问题，我们引入了一个与分类器协同工作的判别器，以产生具有情绪区分性和领域不变性的特征。

具体来说，假设我们分别从源领域和目标领域得到两个数据集 $X = \{X_1, ..., X_m\}$ 和 $X = \{X_1, ..., X_n\}$，其中 M 和 M 分别表示源数据集和目标数据集的数量。为了减轻领域差异，我们引入以下损失函数

L(X, X, u, u) = XM i¼1 logP ð0jXÞ XM j¼1 logP ð1jXÞ; (21)

其中 P ð0jXÞ表示 X 属于源域的概率，而 P ð1jXÞ表示 X 属于目标域的概率，分别地，udenotes 表示判别器的参数。通过最大化上述判别器的损失函数，特征提取过程会产生域不变特征，以缓解情感识别中的域差异。

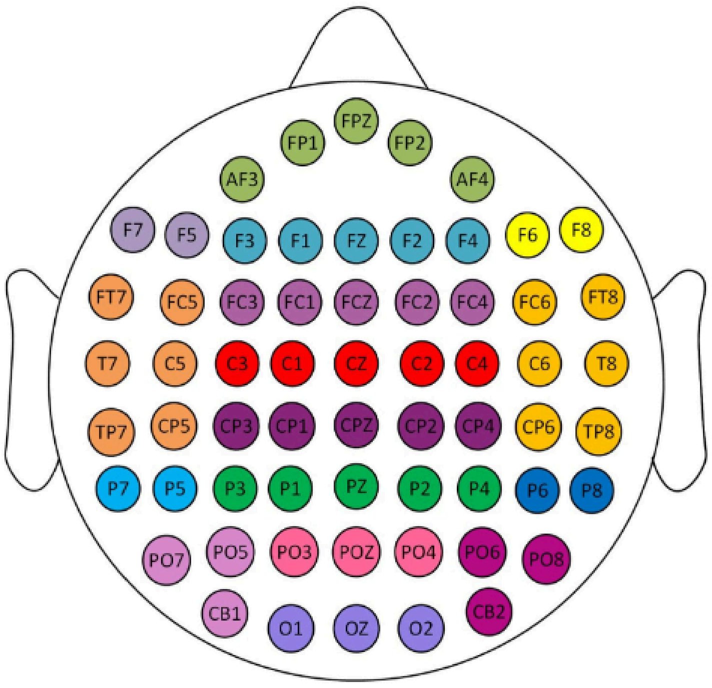

Fig. 3. An illustration of the divisions of the 62 electrodes into 16 clusters, where the same color denotes the electrodes are grouped into the same brain region.

TABLE 1
The EEG Electrodes Associated with Each Brain Region and the Data Size in Each Brain Region in the Experiment

Brain region	Electrode name	EEG data size ($d \times n_j$)
Pre-Frontal	AF3, FP1 FPZ, FP2, AF4	5×5
Frontal	F3, F1 FZ, F2, F4	5×5
Left Frontal	F7, F5	5×2
Right Frontal	F8, F6	5×2
Left Temporal	FT7, FC5, T7 C5, TP7, CP5	5×6
Right Temporal	FT8, FC6, T8 C6, TP8, CP6	5×6
Frontal Central	FC3, FC1 FCZ, FC2, FC4	5×5
Central	C3, C1, CZ, C2, C4	5×5
Central Parietal	CP3, CP1 CPZ, CP2, CP4	5×5
Left Parietal	P7, P5	5×2
Right Parietal	P8, P6	5×2
Parietal	P3, P1, PZ, P2, P4	5×5
Left Parietal Occipital	PO7, PO5, CB1	5×3
Right Parietal Occipital	PO8, PO6, CB2	5×3
Parietal Occipital	PO3, POZ, PO4	5×3
Occipital	O1, OZ, O2	5×3

3.4 The Optimization of R2G-STNN

In the aforementioned section, we pointed out that by minimizing the loss function of (18) we would achieve the better emotion class prediction for the training data samples, while by maximizing the loss function of (21) we would be able to achieve the domain-invariable features to alleviate the domain difference in emotion recognition. Consequently, by simultaneously minimizing the loss function of (18) and maximizing the loss function of (21), we would be able to achieve better emotion classification. For this reason, we define the overall loss function of R2G-STNN as

$$\begin{aligned} L(\mathbf{X}^S, \mathbf{X}^T; \theta_f, \theta_c, \theta_d) = L_c(\mathbf{X}^S; \theta_f, \theta_c) \\ - L_d(\mathbf{X}^S, \mathbf{X}^T; \theta_f, \theta_d), \end{aligned} \tag{22}$$

and our target is to find the optimal parameters to minimize the loss function $L(\mathbf{X}^S, \mathbf{X}^T; \theta_f, \theta_c, \theta_d)$.

The optimal parameters of $L(\mathbf{X}^S, \mathbf{X}^T; \theta_f, \theta_c, \theta_d)$ can be solved by iteratively minimizing $L_c(\mathbf{X}^S; \theta_f, \theta_c)$ and maximizing $L_d(\mathbf{X}^S, \mathbf{X}^T; \theta_f, \theta_d)$. Specifically, we adopt stochastic gradient descent (SGD) algorithm [29] to find the optimal model parameters, i.e.,

$$(\hat{\theta}_f, \hat{\theta}_c) = \arg\min_{\theta_f, \theta_c} L_c(\mathbf{X}^S; (\theta_f, \theta_c), \hat{\theta}_d), \tag{23}$$

$$\hat{\theta}_d = \arg\max_{\theta_d} L_d(\mathbf{X}^S, \mathbf{X}^T; (\hat{\theta}_f, \hat{\theta}_c), \theta_d). \tag{24}$$

Through minimizing the loss function L_c, the feature extractor will able to learn the emotion discriminative features. On the other hand, by maximizing the loss function L_d, it can extract domain-invariant features. Consequently, by simultaneously minimizing L_c while maximizing L_d, we can finally obtain the emotion-discriminative while domain-invariant features for emotion recognition.

Additionally, in solving the optimal parameters of R2G-STNN, we also introduce a gradient reverse layer (GRL) for the discriminator to change the maximizing problem into a minimizing problem, such that the parameters can be optimized by using SGD approach, where GRL acts as an identity transform in the forward-propagation but reverses the gradient sign while performing the back-propagation operation. In this case, the update of the parameters can be formulated as

$$\theta_d \leftarrow \theta_d - \alpha \frac{\partial L_d}{\partial \theta_d}, \quad \theta_f \leftarrow \theta_f + \alpha \frac{\partial L_d}{\partial \theta_f}, \tag{25}$$

where α is the learning rate.

4 Experiments

In this section, we will conduct extensive experiments on SEED database [7] to evaluate the proposed R2G-STNN method. The SEED database was built by using 62-channels' ESI NeuroScan system to record EEG signals, in which the electrodes are located according to the 10-20 system [21]. We group the 62 electrodes into 16 clusters, i.e., the cluster number $N = 16$, based on the spatial locations of the electrodes. Fig. 3 shows the divisions of the 62 electrodes into 16 clusters whereas Table 1 summarizes the EEG electrodes in each cluster (brain region) as well as the handcraft EEG feature set size in each brain region. In the SEED database, there are 15 subjects and each subject contains the EEG data recorded from three sessions. For each subject, every session contains 15 trials of EEG samples and each trial contains 185-238 samples covering three emotion classes, i.e., positive, negative and neutral emotions. As a result, there are totally 3,200 samples in every session, in which each emotion contains about 1,060 samples. Additionally, for each sample, the number of EEG segments T is fixed at $T = 9$ such that each EEG sample corresponds to a $5 \times 62 \times 9$ handcraft feature tensor. In addition, the parameters of d_r, d_g, d_{rt}, and d_{gt} are respectively fixed at 100, 150, 200, and 250 throughout the experiments.

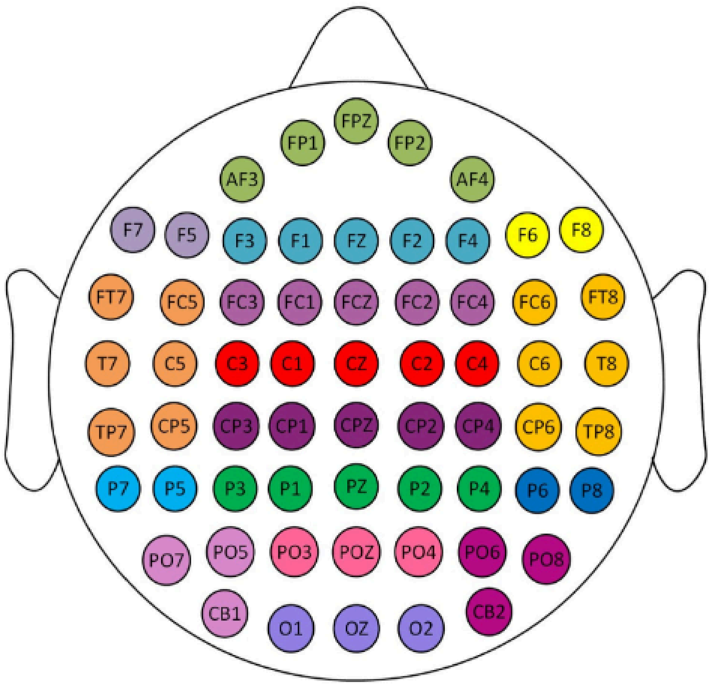

图 3. 62 个电极分为 16 个簇的示意图，其中相同颜色表示电极被分组到同一脑区。

3.4 R2G-STNN 的优化

在上述部分，我们指出通过最小化(18)的损失函数，我们可以实现训练数据样本更好的情感类别预测，而通过最大化(21)的损失函数，我们可以实现域不变特征，以缓解情感识别中的域差异。因此，通过同时最小化(18)的损失函数和最大化(21)的损失函数，我们可以实现更好的情感分类。为此，我们将 R2G-STNN 的整体损失函数定义为

LðX; X; u; u; uÞ ¼ LðX; u; uÞ
LðX; X; u; uÞ; (22)

我们的目标是找到最优参数以最小化损失函数 LðX; X; u; u; uÞ。LðX; X; u; u; uÞ 的最优参数可以通过迭代最小化 LðX; u; uÞ 并最大化 LðX; X; u; uÞ 来求解。具体来说，我们采用随机梯度下降（SGD）算法 [29] 来找到最优模型参数，即，

ð^u; ^uÞ ¼ arg minLðX; ðu; uÞ; ^uÞ; (23)
uf ;uc

û¼ arg max LðX; X; ð^u; ^uÞ; uÞ: (24)
ud

通过最小化损失函数 L，特征提取器将能够学习情感判别特征。另一方面，通过最大化损失函数 L，它可以提取领域不变特征。因此，通过同时最小化 L 并最大化 L，我们最终可以获得用于情感识别的情感判别性同时领域不变特征。

此外，在解决 R2GSTNN 的最优参数时，我们还为判别器引入了一个梯度反转层（GRL），以将最大化问题转换为最小化问题，从而可以使用 SGD 方法优化参数

表 1 实验中与每个脑区相关的脑电图电极以及每个脑区的数据量

脑区	电极名称	脑电图数据大小 ðd nÞ
前额叶	AF3, FP1 FPZ, FP2, AF4	5 5
额叶	F3, F1 FZ, F2, F4	5 5
左额叶	F7, F5	5 2
右侧额叶	F8, F6	5 2
左颞叶	FT7, FC5, T7 C5, TP7, CP5	5 6
右颞叶	FT8, FC6, T8 C6, TP8, CP6	5 6
额叶中央	FC3, FC1 FCZ, FC2, FC4	5 5
中央	C3, C1, CZ, C2, C4	5 5
中央顶叶	CP3, CP1 CPZ, CP2, CP4	5 5
左顶叶	P7, P5	5 2
右顶叶	P8, P6	5 2
顶叶	P3, P1, PZ, P2, P4	5 5
左侧顶叶枕叶	PO7, PO5, CB1	5 3
右顶枕叶	PO8, PO6, CB2	5 3
顶叶-枕叶	PO3, POZ, PO4	5 3
枕叶	O1, OZ, O2	5 3

GRL 在正向传播中充当恒等变换，但在反向传播操作中反转梯度符号。在这种情况下，参数的更新可以表示为

u u a @L ; u uþ a @L ; (25)
@u @u

其中 a 是学习率。

4 实验

在本节中，我们将对 SEED 数据库[7]进行大量实验，以评估所提出的 R2G-STNN 方法。SEED 数据库是使用 62 通道的 ESI NeuroScan 系统记录脑电图信号的，其中电极根据 10-20 系统[21]进行定位。我们根据电极的空间位置将 62 个电极分组为 16 个簇，即簇数 N=16。图 3 显示了 62 个电极被分为 16 个簇的情况，而表 1 则总结了每个簇（脑区）中的脑电图电极以及每个脑区的手工脑电图特征集大小。在 SEED 数据库中，有 15 名受试者，每位受试者包含三个会话记录的脑电图数据。对于每位受试者，每个会话包含 15 次脑电图样本试验，每次试验包含 185-238 个样本，涵盖三种情绪类别，即积极、消极和中性情绪。因此，每个会话共有 3200 个样本，其中每种情绪包含约 1060 个样本。此外，对于每个样本，脑电图片段数 T 固定为 T=9，因此每个脑电图样本对应一个 5×62×9 的手工特征张量。 此外，在实验中，参数 d、d、d 和 d 分别固定在 100、150、200 和 250。

TABLE 2
The Mean Accuracies (ACC) and Standard Deviations (STD) of the Various EEG Emotion Recognition Methods in the Subject-Dependent Experiment

Method	SVM [30]	RF [31]	CCA [32]	GSCCA [4]	DBN [7]	GRSLR [9]
ACC/STD (%)	83.99/9.72	78.46/11.77	77.63/13.21	82.96/9.95	86.08/8.34	87.39/8.64
Method	GCNN [33]	DGCNN [14]	DANN [34]	BiDANN [15]	R2G-STNN	
ACC/STD (%)	87.40/9.20	90.40/8.49	91.36/8.30	92.38/7.04	**93.38/5.96**	

TABLE 3
The Mean Accuracies (ACC) and Standard Deviations (STD) of the Various EEG Emotion Recognition Methods with Respect to Five Frequency Bands in the Subject-Dependent Experiment

Methods	The results (%) of ACC/STD				
	δ	θ	α	β	γ
SVM [30]	60.50 (14.14)	60.95 (10.20)	66.64 (14.41)	80.76(11.56)	79.56(11.38)
RF [31]	64.56(8.32)	65.27(11.64)	65.67 (13.94)	73.35 (14.35)	74.48 (12.80)
CCA [32]	55.30 (12.02)	55.75 (10.99)	64.96 (12.05)	69.16 (11.45)	70.67 (14.06)
GSCCA [4]	63.92 (11.16)	64.64 (10.33)	70.10 (14.76)	76.93 (11.00)	77.98 (10.72)
DBN [7]	64.32 (12.45)	60.77 (10.42)	64.01 (15.97)	78.92 (12.48)	79.19 (14.58)
GRSLR [9]	63.90 (11.83)	62.61 (10.73)	71.11 (9.04)	81.18 (10.74)	81.91 (10.36)
GCNN [33]	72.75 (10.85)	74.40 (8.23)	73.46 (12.17)	83.24 (9.93)	83.36 (9.43)
DGCNN [14]	74.25 (11.42)	71.52 (5.99)	74.43 (12.16)	83.65 (10.17)	85.73 (10.64)
DANN [34]	72.13 (11.22)	68.75 (7.40)	70.27 (10.84)	83.35 (11.46)	87.89 (11.35)
BiDANN [15]	76.97 (10.95)	75.56 (7.88)	81.03 (11.74)	**89.65 (9.59)**	88.64 (9.46)
R2G-STNN	**77.76 (9.92)**	**76.17 (7.43)**	**82.30 (10.21)**	88.35 (10.52)	**88.90 (9.97)**

4.1 Subject-Dependent EEG Emotion Recognition Experiment

In this experiment, we adopt the similar subject-dependent EEG emotion recognition protocol used in [7] and [15] to evaluate the proposed method, where both training and testing data come from the same subject but different EEG trials. Specifically, we choose 9 trials of EEG signals in every session to serve as training data set and use the other 6 trials from the same session as testing data. Then, we use the proposed R2G-STNN method to conduct the EEG emotion recognition experiments and calculate the average recognition accuracy and standard deviation of all the 15 subjects as the final recognition result. For comparison purpose, we also conduct the same experiments using several commonly used methods, which are listed as follows:

- Three baseline methods: Support Vector Machine (SVM) [30], Random Forest (RF) [31], and Canonical Correlation Analysis (CCA) [32];
- Two subspace learning methods: Group Sparse Canonical Correlation Analysis (GSCCA) [8] and Graph Regularized Sparse Linear Regression (GRSLR) [9];
- Five deep learning methods: Deep Belief Networks (DBN) [7], Graph Convolutional Neural Networks (GCNN) [33], Dynamical Graph Convolutional Neural Networks (DGCNN) [14], Domain Adversarial Neural Network (DANN) [34], and Bi-hemisphere Domain Adversarial Neural Network (BiDANN) [15].

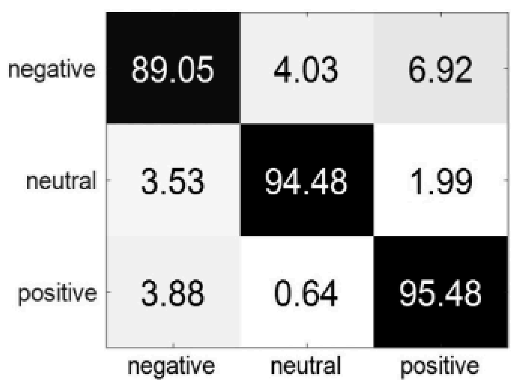

Fig. 4. The EEG emotion recognition confusion matrix of the R2G-STNN method in the subject-dependent experiment.

Table 2 shows the experimental results of the various methods, from which we can see that the proposed R2G-STNN method achieves the recognition accuracy as high as 93.38 percent, which is the best recognition result among the various recognition methods. The better recognition results of R2G-STNN may largely attribute to the fact that R2G-STNN not only utilizes the temporal information of the EEG signal but also the spatial information of the brain regions, which would be in favor of the learning of more discriminative features related to emotion. Moreover, to visualize the confusion among the three emotions, we also depict the confusion matrix of the emotion recognition results in Fig. 4, from which we observe that both positive and neutral emotions are less confused than the negative emotion.

In addition, to investigate the impacts of different frequency bands of EEG signals on the emotion recognition, we also conduct additional experiments by adopting the similar approach used in [8]. Specifically, we first extract DE features from the raw EEG signals with respect to five different frequency bands, i.e., $\delta, \theta, \alpha, \beta$ and γ frequency bands. Then, we conduct EEG emotion recognition experiments based on the DE features extracted from the five frequency bands, respectively. The EEG emotion recognition results are shown in Table 3, from which we can observe that the higher frequency bands such as β and γ frequency bands, achieve

表 2 主体依赖实验中各种脑电图情绪识别方法的平均准确率（ACC）和标准差（STD）

方法	SVM [30]	RF [31]	CCA [32]	GSCCA [4]	DBN [7]	GRSLR [9]
ACC/STD (%)	83.99/9.72	78.46/11.77	77.63/13.21	82.96/9.95	86.08/8.34	87.39/8.64
方法	GCNN [33]	DGCNN [14]	DANN [34]	BiDANN [15]	R2G-STNN	
ACC/STD (%)	87.40/9.20	90.40/8.49	91.36/8.30	92.38/7.04	93.38/5.96	

表 3 在受试者相关实验中，各种脑电图情绪识别方法在五个频段上的平均准确率（ACC）和标准差（STD）

方法	ACC/STD 的结果 (%)				
	d	u	a	b	g
SVM [30]	60.50 (14.14)	60.95 (10.20)	66.64 (14.41)	80.76(11.56)	79.56(11.38)
RF [31]	64.56(8.32)	65.27(11.64)	65.67 (13.94)	73.35 (14.35)	74.48 (12.80)
CCA [32]	55.30 (12.02)	55.75 (10.99)	64.96 (12.05)	69.16 (11.45)	70.67 (14.06)
GSCCA [4]	63.92 (11.16)	64.64 (10.33)	70.10 (14.76)	76.93 (11.00)	77.98 (10.72)
DBN [7]	64.32 (12.45)	60.77 (10.42)	64.01 (15.97)	78.92 (12.48)	79.19 (14.58)
GRSLR [9]	63.90 (11.83)	62.61 (10.73)	71.11 (9.04)	81.18 (10.74)	81.91 (10.36)
GCNN [33]	72.75 (10.85)	74.40 (8.23)	73.46 (12.17)	83.24 (9.93)	83.36 (9.43)
DGCNN [14]	74.25 (11.42)	71.52 (5.99)	74.43 (12.16)	83.65 (10.17)	85.73 (10.64)
DANN [34]	72.13 (11.22)	68.75 (7.40)	70.27 (10.84)	83.35 (11.46)	87.89 (11.35)
BiDANN [15]	76.97 (10.95)	75.56 (7.88)	81.03 (11.74)	89.65 (9.59)	88.64 (9.46)
R2G-STNN	77.76 (9.92)	76.17 (7.43)	82.30 (10.21)	88.35 (10.52)	88.90 (9.97)

4.1 基于被试的脑电情绪识别实验

在本实验中，我们采用与文献[7]和[15]中使用的类似受试者相关脑电图情绪识别协议来评估所提出的方法，其中训练和测试数据均来自同一受试者但不同的脑电图试验。具体而言，我们选择每场会话中的 9 个脑电图信号试验作为训练数据集，并使用同一场会话中的其余 6 个试验作为测试数据。然后，我们使用所提出的 R2G-STNN 方法进行脑电图情绪识别实验，并计算所有 15 个受试者的平均识别准确率和标准差作为最终识别结果。为进行比较，我们还使用几种常用方法进行了相同的实验，具体如下：

三个基线方法：支持向量机（SVM）[30]、随机森林（RF）[31]和典型相关分析（CCA）[32]； 两个子空间学习方法：分组稀疏典型相关分析（GSCCA）[8]和图正则化稀疏线性回归（GRSLR）[9]；

五种深度学习方法：深度信念网络（DBN）[7]、图卷积神经网络（GCNN）[33]、动态图卷积神经网络（DGCNN）[14]、领域对抗神经网络（DANN）[34]和双半球领域对抗神经网络（BiDANN）[15]。表 2 展示了各种方法的实验结果，从中我们可以看到，所提出的 R2GSTNN 方法达到了 93.38%的识别准确率，这是各种识别方法中的最佳识别结果。R2G-STNN 更好的识别结果可能很大程度上归因于 R2G-STNN 不仅利用了脑电信号的时序信息，还利用了脑区的空间信息，这有利于学习与情绪相关的更具判别性的特征。此外，为了可视化三种情绪之间的混淆情况，我们在图 4 中描绘了情绪识别结果的混淆矩阵，从中我们观察到，正面情绪和中性情绪的混淆程度低于负面情绪。

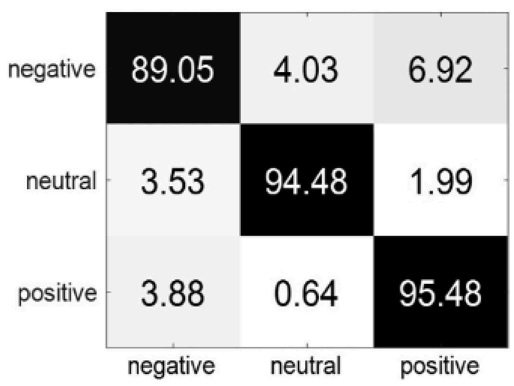

图 4. R2G-STNN 方法在受试者相关实验中的脑电图情绪识别混淆矩阵

此外，为了研究不同频段脑电信号对情绪识别的影响，我们还采用了与[8]中类似的方法进行了额外实验。具体来说，我们首先从原始脑电信号中提取关于五个不同频段的 DE 特征，即 d、u、a、b 和 g 频段。然后，我们分别基于从五个频段提取的 DE 特征进行脑电情绪识别实验。脑电情绪识别结果如表 3 所示，从中我们可以观察到，b 和 g 等高频段取得了

TABLE 4
The Mean Accuracies (ACC) and Standard Deviations (STD) of the Various EEG Emotion Recognition Methods in the Subject-Independent Experiment

Method	KLIEP [36]	ULSIF [37]	STM [38]	SVM [30]	KPCA [39]
ACC/STD (%)	45.71/17.76	51.18/13.57	51.23/14.82	56.73/16.29	61.28/14.62
Method	TCA [40]	TKL [41]	SA [42]	GFK [43]	T-SVM [44]
ACC/STD(%)	63.64/14.88	63.54/15.47	69.00/10.89	71.31/14.09	72.53/14.00
Method	TPT [45]	DGCNN [14]	DANN [34]	BiDANN [15]	R2G-STNN
ACC/STD (%)	76.31/15.89	79.95/9.02	75.08/11.18	83.28/9.60	**84.16/7.63**

TABLE 5
The Mean Accuracies (ACC) and Standard Deviations (STD) of the Various EEG Emotion Recognition Methods with Respect to Five Frequency Bands in the Subject-Independent Experiment

Methods	The results (%) of ACC (STD)				
	δ	θ	α	β	γ
KLIEP [36]	39.22 (11.31)	35.98 (7.50)	33.31 (6.60)	44.47 (12.89)	42.05 (12.65)
ULSIF [37]	41.32 (11.30)	36.27 (6.84)	38.94 (8.30)	41.87 (13.64)	41.02 (11.65)
STM [38]	44.16 (9.60)	40.89 (8.22)	40.37 (9.82)	42.09 (13.34)	47.97 (12.43)
SVM [30]	43.06 (8.27)	40.07 (6.50)	43.97 (10.89)	48.63 (10.29)	51.59 (11.83)
TCA [40]	44.10 (8.22)	41.26 (9.21)	42.93 (14.33)	43.93 (10.06)	48.43 (9.73)
TKL [41]	48.36 (10.31)	52.60 (11.84)	52.89 (11.07)	55.47 (9.80)	59.81 (12.41)
SA [42]	53.23 (7.47)	50.60 (8.31)	55.06 (10.60)	56.72 (10.78)	64.47 (14.96)
GFK [43]	52.73 (11.90)	54.07 (06.78)	54.98 (11.49)	59.29 (10.75)	66.92 (10.97)
DGCNN [14]	49.79 (10.94)	46.36 (12.06)	48.29 (12.28)	56.15 (14.01)	54.87 (17.53)
DANN [34]	56.66 (6.48)	54.95 (10.45)	59.37 (10.57)	67.14 (7.10)	71.30 (10.84)
BiDANN[15]	62.04 (6.64)	62.13 (7.37)	63.31 (11.46)	73.55 (8.83)	73.25 (9.21)
R2G-STNN	**63.34 (5.31)**	**63.78 (7.53)**	**64.27 (10.88)**	**74.85 (8.02)**	**74.54 (8.41)**

better performance than the lower ones such as δ, θ and α frequency bands. This observation coincides with the neurophysiology research findings about emotion [35].

4.2 Subject-Independent EEG Emotion Recognition Experiment

In this experiment, we will investigate the subject-independent EEG emotion recognition problem, in which the training EEG data samples and the testing ones come from different subjects [15], [46]. To this end, we adopt leave-one-subject-out (LOSO) cross validation strategy to conduct the experiment, in which we circularly take one subject's EEG signals as testing data and the EEG signals of all the other subjects as training data. The average result of all the recognition accuracies is then calculated after each subject has been used once as testing data. For comparison purpose, we also conduct the same experiment using several methods, which are listed as follows:

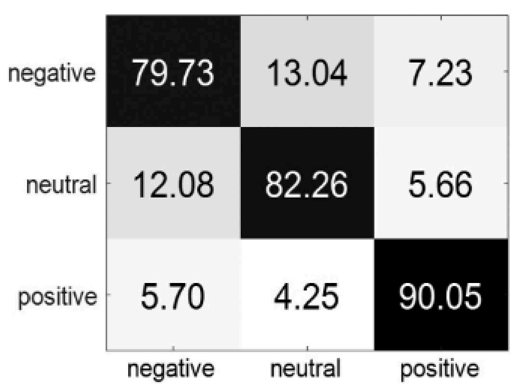

Fig. 5. The EEG emotion recognition confusion matrix of the R2G-STNN method in the subject-independent experiment.

- Two baseline methods: SVM [30] and kernel principal component analysis (KPCA) [39];
- Nine transfer subspace learning methods: Kullback-Leibler importance estimation procedure (KLIEP) [36], unconstrained least-squares importance fitting (ULSIF) [37], selective transfer machine (STM) [38], transfer component analysis (TCA) [40], transfer kernel learning (TKL) [41], subspace alignment (SA) [42], geodesic flow kernel (GFK) [43], transductive SVM (T-SVM) [44], and transductive parameter transfer (TPT) [45];
- Three recent deep learning methods: DGCNN [14], DANN [34] and BiDANN [15].

The experimental results of the various methods are shown in Table 4, from which we can again see that R2G-STNN achieves the state-of-the-art performance among the various methods. To visualize the confusion among the three emotions recognized by R2G-STNN, we depict the confusion matrix according to the recognition results. Fig. 5 shows the results of the confusion matrix, from which we can observe that the positive emotion is less confused than both negative and neutral emotions.

Similar to the subject-dependent experiment, in this experiment we also conduct additional experiments to investigate the impacts of different frequency bands of EEG signals on the emotion recognition. Table 5 shows the experimental results, from which we can also observe that the higher frequency bands achieve better recognition results than the lower ones. Moreover, from Table 5 we can see that R2G-STNN achieves the best recognition results for all the five frequency bands among the various methods.

表 4 各种脑电波情绪识别方法在受试者无关实验中的平均准确率（ACC）和标准差（STD）

方法	KLIEP [36]	ULSIF [37]	STM [38]	SVM [30]	KPCA [39]
ACC/STD (%)	45.71/17.76	51.18/13.57	51.23/14.82	56.73/16.29	61.28/14.62
方法	TCA [40]	TKL [41]	SA [42]	GFK [43]	T-SVM [44]
ACC/STD(%)	63.64/14.88	63.54/15.47	69.00/10.89	71.31/14.09	72.53/14.00
方法	TPT [45]	DGCNN [14]	DANN [34]	BiDANN [15]	R2G-STNN
ACC/STD (%)	76.31/15.89	79.95/9.02	75.08/11.18	83.28/9.60	**84.16/7.63**

表 5 在受试者无关实验中，各种脑电图情绪识别方法在五个频段上的平均准确率（ACC）和标准差（STD）

方法	结果 (%) 的 ACC (STD)				
	d	u	a	b	g
KLIEP [36]	39.22 (11.31)	35.98 (7.50)	33.31 (6.60)	44.47 (12.89)	42.05 (12.65)
ULSIF [37]	41.32 (11.30)	36.27 (6.84)	38.94 (8.30)	41.87 (13.64)	41.02 (11.65)
STM [38]	44.16 (9.60)	40.89 (8.22)	40.37 (9.82)	42.09 (13.34)	47.97 (12.43)
SVM [30]	43.06 (8.27)	40.07 (6.50)	43.97 (10.89)	48.63 (10.29)	51.59 (11.83)
TCA [40]	44.10 (8.22)	41.26 (9.21)	42.93 (14.33)	43.93 (10.06)	48.43 (9.73)
TKL [41]	48.36 (10.31)	52.60 (11.84)	52.89 (11.07)	55.47 (9.80)	59.81 (12.41)
SA [42]	53.23 (7.47)	50.60 (8.31)	55.06 (10.60)	56.72 (10.78)	64.47 (14.96)
GFK [43]	52.73 (11.90)	54.07 (06.78)	54.98 (11.49)	59.29 (10.75)	66.92 (10.97)
DGCNN [14]	49.79 (10.94)	46.36 (12.06)	48.29 (12.28)	56.15 (14.01)	54.87 (17.53)
DANN [34]	56.66 (6.48)	54.95 (10.45)	59.37 (10.57)	67.14 (7.10)	71.30 (10.84)
BiDANN[15]	62.04 (6.64)	62.13 (7.37)	63.31 (11.46)	73.55 (8.83)	73.25 (9.21)
R2G-STNN	63.34 (5.31)	63.78 (7.53)	64.27 (10.88)	74.85 (8.02)	74.54 (8.41)

比 d、u 和 a 频段等较低频段表现更好。这一观察结果与关于情绪的神经生理学研究发现相吻合[35]。

4.2 受试者无关的脑电情绪识别实验

在本实验中，我们将研究受试者无关的脑电图情绪识别问题，其中训练的脑电图数据样本和测试样本来自不同的受试者[15][46]。为此，我们采用留一受试者交叉验证策略进行实验，其中我们循环地取一个受试者的脑电图信号作为测试数据，将所有其他受试者的脑电图信号作为训练数据。在每次一个受试者被用作测试数据后，计算所有识别准确率的平均结果。为了比较，我们还使用几种方法进行了相同的实验，这些方法列如下：

两种基线方法：支持向量机（SVM）[30]和核主成分分析（KPCA）[39]；九种迁移子空间学习方法：Kullback-Leibler 重要性估计程序（KLIEP）[36]、无约束最小二乘重要性拟合（ULSIF）[37]、选择性迁移机（STM）[38]、迁移成分分析（TCA）[40]、迁移核学习（TKL）[41]、子空间对齐（SA）[42]、测地流核（GFK）[43]、判别支持向量机（T-SVM）[44]和判别参数迁移（TPT）[45]；三种最新的深度学习方法：DGCNN [14]、DANN [34]和 BiDANN [15]。各种方法的实验结果如表 4 所示，从中我们可以再次看到 R2GSTNN 在各种方法中实现了最先进的性能。为了可视化 R2G-STNN 识别的三种情绪之间的混淆情况，我们根据识别结果绘制了混淆矩阵。图 5 显示了混淆矩阵的结果，从中我们可以观察到，积极情绪的混淆程度低于消极和中性情绪。

与受试者相关的实验类似，在这个实验中，我们也进行额外的实验来研究脑电波信号不同频段对情绪识别的影响。表 5 显示了实验结果，从中我们也可以观察到，高频段比低频段取得了更好的识别结果。此外，从表 5 中我们可以看到，在各种方法中，R2G-STNN 在五个频段中均实现了最佳的识别结果。

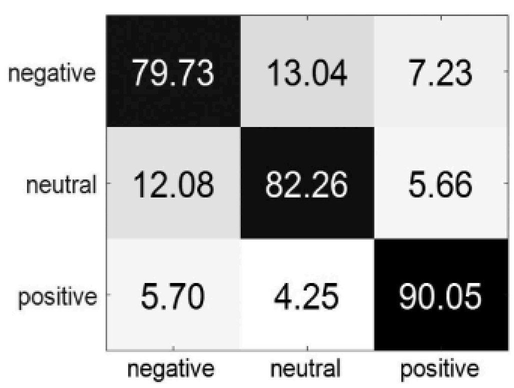

图 5. R2G-STNN 方法在受试者无关实验中的脑电图情绪识别混淆矩阵。

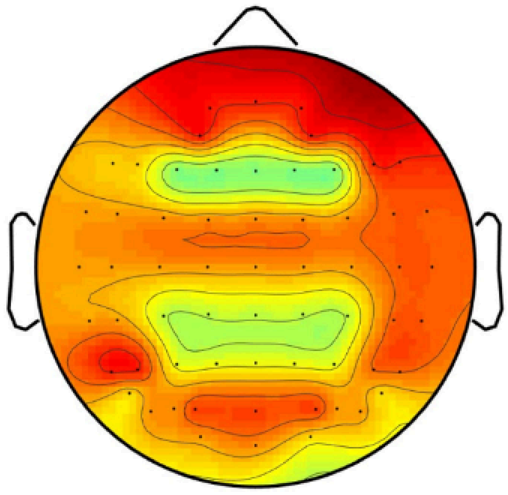

Fig. 6. The topographical map of the sum of each row of **W**, where deeper red color denotes more significant contribution of the corresponding brain region.

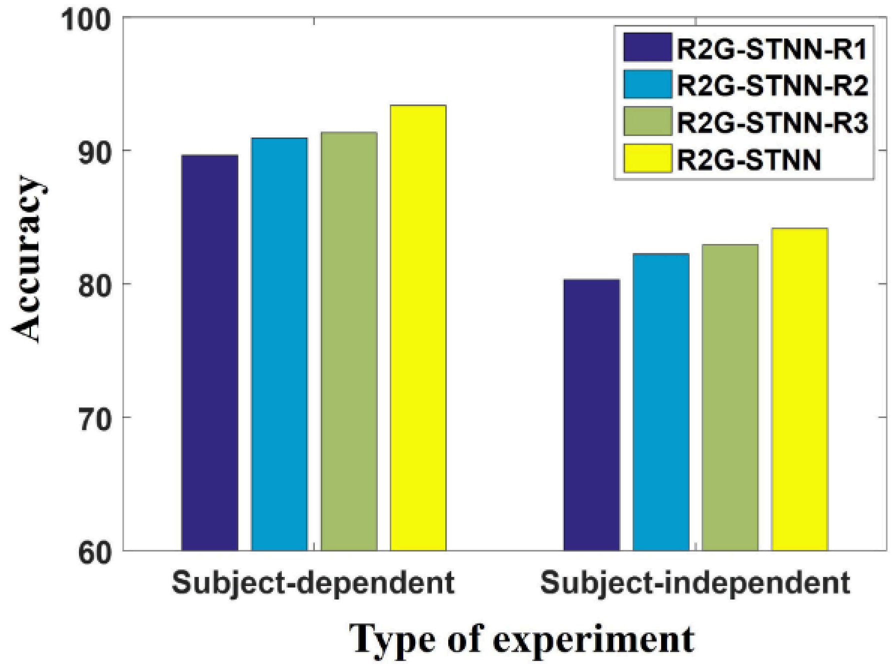

Fig. 7. The comparison of EEG emotion recognition results among four methods: (1) R2G-STNN-R1; (2) R2G-STNN-R2; (3) R2G-STNN-R3; (4) R2G-STNN.

To visualize the impact of the different brain regions on emotion recognition, we also depict the weight distribution over brain regions based on the weighted matrix **W** defined in (5), in which the sum of each row of **W** is used to demonstrate the contribution of the corresponding brain region to the emotion recognition. Fig. 6 shows the topographical map of the sum of each row of **W**, in which the areas with deeper red color mean significant contributions of the corresponding brain regions. From Fig. 6, we can see that the frontal brain regions are important in EEG emotion recognition experiment, which coincides with the cognition observations of biological psychology [47].

5 Discussions and Conclusions

In this paper, we proposed a novel R2G-STNN method inspired by the neuroscience findings that the human brain regions have different responses to emotion and applied it to EEG emotion recognition, in which a hierarchical feature learning procedure from regional brain to global brain is proposed to extract the spatial-temporal EEG features. Extensive experiments on SEED EEG emotional database demonstrated that the proposed R2G-STNN method achieves the state-of-the-art performance in both subject-dependent and subject-independent EEG emotion recognition. The better recognition performance of R2G-STNN may largely attribute to the fact the R2G-STNN makes use of weighting layer and both regional and global temporal layers. To verify these points, we also conduct additional experiments by removing some of the layers, which include the following three reduced models:

1) R2G-STNN-R1, which removes both dynamic weighting layer and regional temporal feature;
2) R2G-STNN-R2, which neglects the regional temporal feature and only uses the global temporal feature as the final vector for classification;
3) R2G-STNN-R3, which treats all the brain regions equally, but has the same spatial-temporal structure with R2G-STNN.

The experimental results are shown in Fig. 7, from which we can see that, for both subject-dependent and subject-independent experiments, the recognition accuracies of R2G-STNN-R1, R2G-STNN-R2, R2G-STNN-R3, and R2G-STNN have the following relationship:

$$\begin{aligned} \text{R2G-STNN-R1} &< \text{R2G-STNN-R2} \\ &< \text{R2G-STNN-R3} < \text{R2G-STNN}. \end{aligned} \tag{26}$$

The relationship in (26) demonstrates the importance of the dynamic weighting layer and regional temporal feature learning layer, each of which can improve the classification performance in both subject-dependent and subject-independent EEG emotion recognition experiments.

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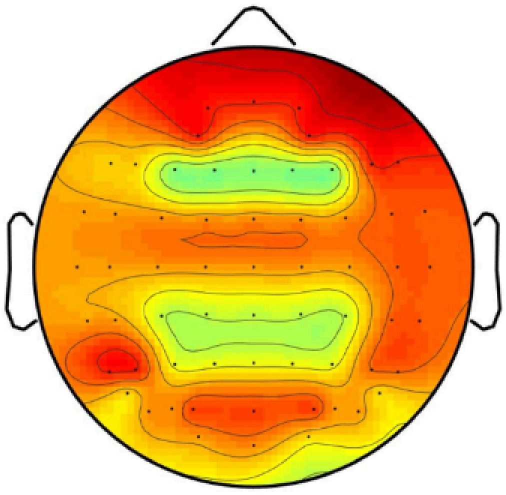

图 6. W 的每一行之和的拓扑图，其中更深的红色表示相应脑区的贡献更显著。

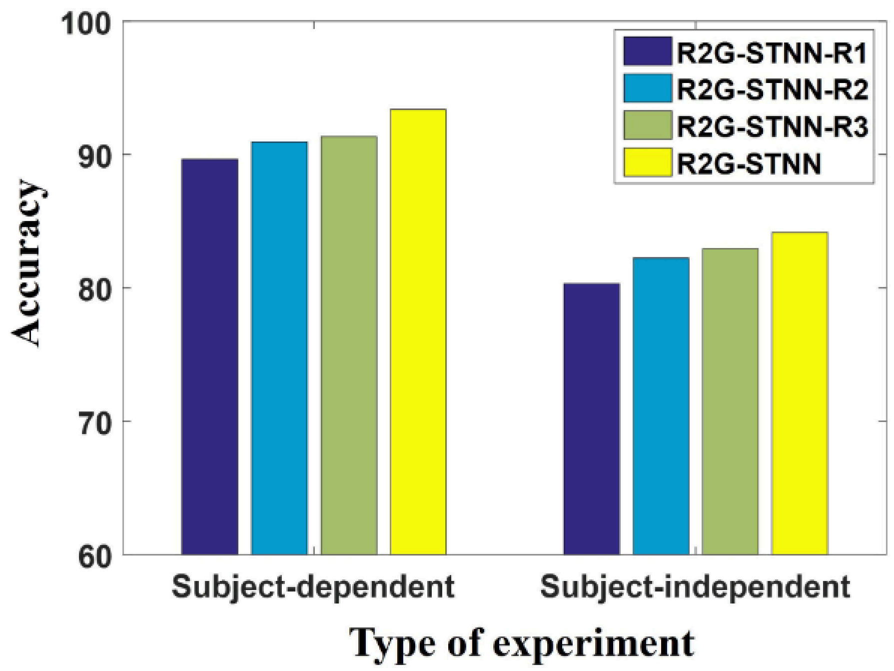

图 7. 四种方法在脑电图情绪识别结果上的比较：(1) R2G-STNN-R1；(2) R2G-STNN-R2；(3) R2G-STNN-R3；(4) R2G-STNN。

为了可视化不同脑区对情绪识别的影响，我们还基于在(5)中定义的加权矩阵 W，描绘了脑区间的权重分布，其中 W 的每一行的和被用来展示相应脑区对情绪识别的贡献。图 6 显示了 W 的每一行的和的拓扑图，其中颜色越深表示相应脑区的贡献越显著。从图 6 中，我们可以看到额叶脑区在脑电图情绪识别实验中非常重要，这与生物心理学中的认知观察结果一致[47]。

5 讨论与结论

在本文中，我们提出了一种受神经科学发现启发的全新 R2G-STNN 方法，该方法认为人脑不同区域对情绪的反应不同，并将其应用于脑电图情绪识别，其中提出了一种从区域脑到全局脑的层次化特征学习过程，以提取脑电图的时空特征。在 SEED 脑电情绪数据库上的大量实验表明，所提出的 R2G-STNN 方法在受试者相关和受试者无关的脑电图情绪识别方面均达到了当前最佳性能。

R2G-STNN 的更好识别性能可能很大程度上归因于 R2G-STNN 使用了加权层以及区域和全局时间层。为了验证这些点，我们还通过移除一些层进行了额外的实验，包括以下三个简化模型：

1) R2G-STNN-R1，该模型移除了动态加权层和区域时间特征；

2) R2G-STNN-R2，该模型忽略了区域时间特征，仅使用全局时间特征作为分类的最终向量；

3) R2G-STNN-R3，该模型对所有脑区一视同仁，但与 R2G-STNN 具有相同的时空结构。

实验结果如图 7 所示，从中我们可以看到，对于受试者相关和受试者无关的实验，R2GSTNN-R1、R2G-STNN-R2、R2G-STNN-R3 和 R2G-STNN 的识别准确率具有以下关系：

$$\text{R2G-STNN-R1} < \text{R2G-STNN-R2} < \text{R2G-STNN-R3} < \text{R2G-STNN}: \quad (26)$$

公式(26)展示了动态加权层和区域时间特征学习层的重要性，每个层都能在受试者相关和受试者无关的脑电图情绪识别实验中提升分类性能。

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Yang Li received the BS degree in electronic information and science technology from the School of Physics and Electronics, Shandong Normal University, China, in 2012, the MS degree in electronic and communication engineering from the School of Electronic Engineering, Xidian University, China, in 2015. Currently, he is working toward the PhD degree in information and communication engineering at Southeast University, China. His researches focus on Pattern Recognition and Machine Learning. He is a student member of the IEEE.

Wenming Zheng (SM'18) received the BS degree in computer science from Fuzhou University, Fuzhou, China, in 1997, the MS degree in computer science from Huaqiao University, Quanzhou, China, in 2001, and the PhD degree in signal processing from Southeast University, Nanjing, China, in 2004. Since 2004, he has been with the Research Center for Learning Science, Southeast University, where he is currently a professor with the School of Biological Science and Medical Engineering and the Key Laboratory of Child Development and Learning Science of the Ministry of Education. His current research interests include affective computing, pattern recognition, machine learning, and computer vision. He served as an associate editor of several peer-reviewed journals, such as *IEEE Transactions on Affective Computing*, *Neurocomputing*, and the *Visual Computer*. He is a senior member of the IEEE.

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以及情绪的调节者,"生物心理学，第 67 卷，第 1/2 期，第 7-50 页，2004.

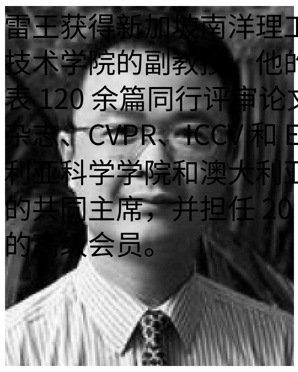
杨立于 2012 年获得山东师范大学物理与电子学院电子信息科学技术专业的学士学位，2015 年获得西安电子科技大学电子工程学院电子与通信工程专业的硕士学位。目前，他正在中国东南大学攻读信息与通信工程专业的博士学位。他的研究方向集中在模式识别和机器学习。他是 IEEE 的学生会员。

郑文明（SM'18）于 1997 年在福州大学获得计算机科学学士学位，2001 年在华侨大学获得计算机科学硕士学位，2004 年在东南大学获得信号处理博士学位。自 2004 年起，他在东南大学学习科学研究中心工作，目前是生物科学与医学工程学院教授，同时也是教育部儿童发展与学习科学重点实验室的成员。他的当前研究兴趣包括情感计算、模式识别、机器学习和计算机视觉。他曾担任 IEEE Transactions on Affective Computing、Neurocomputing 和 Visual Computer 等同行评审期刊的副编辑。他是 IEEE 的高级会员。

Lei Wang received the PhD degree from Nanyang Technological University, Singapore. He is now associate professor at School of Computing and Information Technology of University of Wollongong, Australia. His research interests include machine learning, pattern recognition, and computer vision. He has published more than 120 peer-reviewed papers, including those in highly regarded journals and conferences such as the *IEEE Transactions on Pattern Analysis and Machine Intelligence*, the *International Journal of Computer Vision*, CVPR, ICCV and ECCV, etc. He was awarded the Early Career Researcher Award by Australian Academy of Science and Australian Research Council. He served as the general co-chair of DICTA 2014 and on the Technical Program Committees of more than 20 international conferences and workshops. He is senior member of the IEEE.

Yuan Zong received the BS and MS degrees in electronics engineering from Nanjing Normal University, Nanjing, China, in 2011 and 2014, respectively. He is currently working toward the PhD degree in the Key Laboratory of Child Development and Learning Science of Ministry of Education, School of Biological Sciences and Medical Engineering, Southeast University, Nanjing, China. From 2016 to 2017, he was working as a visiting student with the Center for Machine Vision and Signal Analysis, University of Oulu, Finland. His research interests include affective computing, pattern recognition, and computer vision. He is a student member of the IEEE.

Zhen Cui received the PhD degree from Institute of Computing Technology (ICT), Chinese Academy of Sciences, in 2014. He was a research fellow with the Department of Electrical and Computer Engineering, National University of Singapore (NUS) from 2014 to 2015. He also spent half a year as a research assistant on Nanyang Technological University (NTU) from Jun. 2012 to Dec. 2012. Currently he is a professor of Nanjing University of Science and Technology, China. His research interests cover computer vision, pattern recognition and machine learning, especially focusing on deep learning, manifold learning, sparse coding, face detection/alignment/recognition, object tracking, image super resolution, emotion analysis, etc. He has published several papers in the top conferences NIPS/CVPR/ECCV and some journals of IEEE Transactions. More details can be found in http://aip.seu.edu.cn/zcui/. He is a member of the IEEE.



雷王获得新加坡南洋理工大学博士学位。他现在是澳大利亚伍伦贡大学计算与信息技术学院的副教授。他的研究兴趣包括机器学习、模式识别和计算机视觉。他已发表 120 余篇同行评审论文，包括在 IEEE 模式分析与机器智能汇刊、国际计算机视觉杂志、CVPR、ICCV 和 ECCV 等备受推崇的期刊和会议上发表的论文。他获得了澳大利亚科学学院和澳大利亚研究委员会的青年研究奖。他曾担任 2014 年 DICTA 大会的共同主席，并担任 20 余个国际会议和研讨会的技术程序委员会成员。他是 IEEE 的高级会员。

袁宗于 2011 年和 2014 年分别获得南京师范大学电子工程专业的学士和硕士学位。目前，他在中国教育部儿童发展与学习科学重点实验室，南京东南大学生物科学与医学工程学院攻读博士学位。2016 年至 2017 年，他作为交换生在芬兰奥卢大学的机器视觉与信号分析中心学习。他的研究兴趣包括情感计算、模式识别和计算机视觉。他是 IEEE 的学生会员。

崔振在 2014 年获得中国科学院计算技术研究所（ICT）的博士学位。2014 年至 2015 年，他担任新加坡国立大学（NUS）电气与计算机工程系的研究员。2012 年 6 月至 2012 年 12 月，他还以研究助理的身份在南洋理工大学（NTU）工作了半年。目前，他是中国南京科技大学的教授。他的研究兴趣涵盖计算机视觉、模式识别和机器学习，尤其专注于深度学习、流形学习、稀疏编码、人脸检测/对齐/识别、目标跟踪、图像超分辨率、情感分析等。他在 NIPS/CVPR/ECCV 等顶级会议和 IEEE Transactions 的一些期刊上发表了多篇论文。更多详情请访问 http://aip.seu.edu.cn/zcui/。他是 IEEE 的会员。

" 如需了解更多关于此主题或其他任何计算机相关主题的信息，请访问我们的数字图书馆 www.computer.org/csdl。